EIGSEP Memo 005

Marjum Pass 2026-07: beam reconstruction from the transmitter raster

EIGSEP Team

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Abstract

How box-air's beam is measured from the 07-17 transmitter raster, and how well. The transmitter's comb teeth, seen through box-air as it rolls and tips, sample the beam's response to two orthogonal polarizations over the whole sphere of directions. This memo describes the data selection, the background under each tooth, the beam model (spherical harmonics times a smooth spectral basis, regularized toward the HFSS simulation), the geometry fit and the held-out test, and states how far the reconstructed beam improves on HFSS. The reference product is `derived/beam/empirical_raster_v0017`, fitted on pointing table v5, whose azimuth is left-handed about the boresight (memo 003), with the antenna and transmitter at memo 002's adopted `v0005` posterior means and a tilt of the elevation axle along the highline as a fitted correction. Its DPSS penalty is held at 10^{-2}: at the weaker penalty the validation scores prefer, the exported beam near 66.4 MHz has a lobe 59° from the boresight at 5.2× HFSS's peak, which the held-out scores do not see. This memo does not refit it: every number, figure and table is computed by `make_products.py` in this directory from the product's per-sample data and predictions (Appendix A). Pointing, the mount and the azimuth zero are in memo 003.

1 Summary

- **Data**: 64 transmitter teeth between 64.2 and 240.5 MHz (25 on arm 0, 39 on arm 1), 6552 samples over 20:26:00–21:28:40 UTC on 07-17 at the ∼91m height era, split by 12° azimuth stripes into training (4315), validation (1043) and test (1194) samples.
- **The empirical beam beats HFSS on the test stripes**: fractional rms 0.076 against 0.091 (means over teeth weighted by sample count), better on 44 of 64 teeth, with the worst tooth at 0.250 against 0.270 (Table 6).
- **The residual is structure, not noise**: it is ∼5× the white-noise floor (0.014). What is left is in the model, the background or the geometry.
- **Geometry**: azimuth is left-handed (memo 003). The fit puts the table's zero +15.93° from the highline, as memo 003 finds from the arm ratio alone, and transmitter arm 0 at bearing 172°. The beam cannot be tipped along the highline by the az and el zeros; a tilt of the elevation axle does it. With the transmitter held at memo 002's position the fit wants +4.74°, and box-air's elevation IMU measures +5.97° about the same axis with the same sign. The tilt takes the validation score from 0.093 to 0.076. It replaces the transmitter offsets of the earlier products, which ran to their bound.
- **Change from v0013** (right-handed, transmitter offsets free): test scores 0.091 (HFSS) and 0.076 (DPSS) against 0.111 and 0.083, on different test samples (§4).
- **Penalty**: the validation scores cannot choose the DPSS penalty. At 10^{-3}, where 10^{-2} is 5.9% worse on validation, the exported beam at 66.4 MHz differs from HFSS by 3.8 dB rms on the sampled pixels and its peak, 59° from the boresight, is 5.2× HFSS's. At 10^{-2} these are at most 0.4 dB and 1.0× below 90 MHz (Table 2). The penalty is held at 10^{-2}.
- **Limits**: power-only data leave the beam's phase undetermined; directions the raster never sampled (42% of HEALPix pixels are sampled) are set by the HFSS prior and the penalty, and no test reaches them (§6).

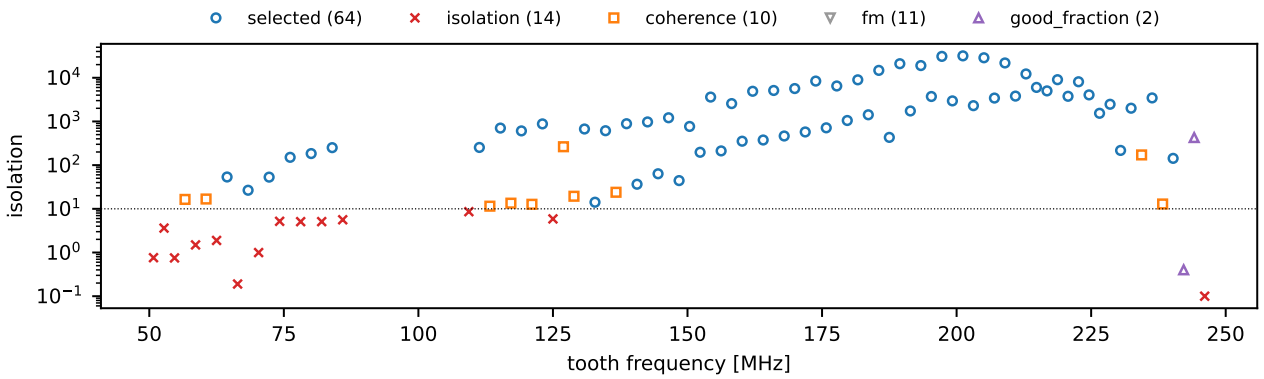


Figure 1: Tooth isolation against frequency, marked by the reason each candidate was kept or dropped. The dotted line is the isolation threshold.

2 Data

Samples. Box-air, antenna-input rows of the elevation-sweep raster, 20:26:00–21:28:40 UTC on 07-17 (memo 003, The 07-17 transmitter raster), with pointing-table v5 quality `ok`: commanded azimuth, left-handed about the boresight, and IMU elevation. Only rows in the ∼91m height era are kept (234 dropped; `curation/mode_table.jsonl`), and isolated elevation-solution glitches are dropped. Throughout the raster box-air rolls about its boresight in 5° steps while the elevation axis turns through full revolutions, so the transmitter is seen at every angle from the boresight and at a range of roll angles.

Background under each tooth. The comb sits on box-air's sky and receiver spectrum. For each integration a robust DPSS model is fitted to the spectrum in 48 MHz windows with a 150 ns delay half-width, masking each tooth's channel and the channels either side, and the FM band (87–108.5 MHz), with 4σ clipping; the tooth signal is the measured power minus this background. Since v0012 the background keeps every DPSS mode with concentration $\geq 10^{-6}$ (21 modes); the earlier floor$(2NW)+1$ rule could not represent a constant near the window edges (`eigsep_data.beam_mapping.tx_background`). Broadband dropouts are masked by a gap-ratio test, and `raster_flags_v0001` removes 242 samples and 32 tooth spikes; the 3 samples it does not cover count as flagged.

Which teeth. The candidates are the 101 channels $\equiv 0 \bmod 8$ inside the HFSS frequency range. A tooth is kept if it is isolated (the 90th percentile of its signal-to-background ratio over the raster is at least 10 times the larger of its background error in a transmitter-off control interval, 17:03–17:54, and its local residual), if at least half its samples are good, and if it is coherent with its neighbours: its log signal must correlate at ≥ 0.9 with the mean of the two nearest selected teeth on the same transmitter arm, iterated by dropping the least coherent tooth (`tooth_selection_v0002`). 14 fail isolation, 11 are in the FM band, 10 fail coherence and 2 have too few good samples (Fig. 1). The coherence test removes teeth with intermittent RFI that pass isolation.

3 Model

Coupling. Box-air is fixed in position relative to the transmitter, so the transmitter's own gain toward box-air is the same for every sample and is absorbed, with the drive and the cable losses, into one positive gain g_f per tooth. The model power for tooth f, driven on transmitter arm $a(f)$, is

$$P_f(t) = g_f \left|\mathbf{E}_f(\theta_t, \phi_t) \cdot \hat{\mathbf{e}}_{a(f)}(t)\right|^2, \tag{1}$$

where (θ_t, ϕ_t) is the transmitter's direction in box-air's body frame (memo 003, Eq. 1), $\hat{\mathbf{e}}_a$ is the field direction of transmitter arm a (horizontal dipoles at polarization angle α and $\alpha + 90°$) expressed in the same frame, and $\mathbf{E}_f$ is box-air's complex vector far-field pattern. Both arms see the same beam through orthogonal polarizations.

The simulated beam. Dominic's HFSS model of box-air's bowtie antenna gives $\mathbf{E}_f$ as Cartesian field components on a HEALPix grid (nside 32) at 52 frequencies, 50.8–250.0 MHz (`eigsep_data/hfss_beam_maps/bowtie_beam.npz`). The fields are normalized to unit mean power at 190 MHz with the common phase removed, and interpolated to the teeth.

The empirical beam. Each Cartesian component is a sum of real spherical harmonics to $\ell_{\max} = 8$ (81 terms) with coefficients that are themselves smooth in frequency, expanded in a spectral basis: $\mathbf{E}_f(\theta, \phi) = \sum_k s_k(f) \sum_{\ell m} \mathbf{C}_{k\ell m} Y_{\ell m}(\theta, \phi)$. Two spectral bases are fitted: 8 DPSS modes with a 12.5 ns delay half-width (sinc-continued across the band), and 5 principal components of the HFSS fields. $\ell_{\max} = 8$ represents the HFSS fields to 0.0039 worst-case relative error, and the two spectral bases to under 1%. The loss is the mean over teeth of the squared residual, each tooth scaled to unit rms, plus a prior $\lambda \|\mathbf{C} - \mathbf{C}_0\|^2 / \|\mathbf{C}_0\|^2$ toward the HFSS beam's own coefficients $\mathbf{C}_0$; the gains are fitted in log so they stay positive (JAX, L-BFGS-B; `tx_fit.JointBeamFit`).

Training, validation and test. Samples are assigned by stripes of commanded azimuth 12° wide: stripe k is validation if $k \bmod 5 = 1$, test if $k \bmod 5 = 3$, and training otherwise. The beam is fitted on training samples at $\lambda = 10^{-2}, 10^{-3}, 10^{-4}$. Validation differences between penalties are about 1%, and the weakest may stop at the iteration limit, so each basis takes the strongest converged penalty whose validation score is within 1% of the best. That rule chooses 0.01 for PCA. For DPSS it chooses 10^{-3}, against which 10^{-2} is 5.9% worse on validation, but the penalty is held at 10^{-2} (Table 1; `fit_beam.py --penalty dpss=1e-2`). In `empirical_raster_v0016`, the same fit at 10^{-3}, the exported beam at 66.4 MHz has a lobe 59° from the boresight at 5.2× HFSS's peak, and its total power differs from HFSS's by 3.8 dB rms even on the sampled pixels (Table 2), yet its held-out score below 87 MHz is the better of the two. The data measure the power in the transmitter's polarization along the raster's track, not the total power or the directions off it, so the scores cannot rule such a beam out; the stronger penalty keeps it near HFSS there. The beam fitted on the training stripes at the chosen penalty is the *held-out* beam, whose test score is an independent measure. The chosen fit is then refitted on all samples and exported in the HFSS file format (*refit* and *exported*; their scores are in-sample, and the export changes the predictions by at most 0.21%). Outside 64.2–240.5 MHz the exported slices are the normalized HFSS beam.

Table 1: DPSS penalty selection: each penalty's validation median per-tooth fractional rms and whether it converged, and the count-weighted test fractional rms by band of the beam fitted on the training stripes at that penalty.

DPSS penalty λ	10^{-2}	10^{-3}	10^{-4}
validation median rms	0.0681	0.0643	0.0659
converged	yes	yes	no
test fractional rms, band [MHz]			
50–87	0.065	0.060	0.060
87–145	0.081	0.082	0.082
145–190	0.072	0.073	0.073
190–235	0.080	0.076	0.075
235–245	0.070	0.063	0.062

4 Geometry

The positions are memo 002's adopted `v0005` posterior means (antenna in the 91 m era, transmitter; `memos/memo-002-marjum-2026-07-geometry/tables/summary.json`, which this memo checks against the product's provenance); the mount and convention are memo 003's, with $\psi = 142.164°$ and the table's azimuth left-handed about the boresight (pointing table v5). Two quantities are not known in advance: the azimuth offset added to the table's azimuth, and the transmitter's arm-0 angle α (the arm-0 field is $(-\sin\alpha, \cos\alpha, 0)$ in ENU, a compass bearing of $180° - \alpha$). A grid search with the HFSS beam finds them together (Fig. 2): offset $+16°$ and $\alpha = 8°$, normalized rms 0.120, against 0.328 at the best α with zero offset.

Table 2: The exported DPSS beam against HFSS at each HFSS frequency below 90 MHz, and the worst above it, for this product (DPSS penalty 10^{-2}) and for `empirical_raster_v0016`, the same fit at the 10^{-3} the selection rule chose. Each total-power map is scaled to unit visit-weighted mean on the sampled pixels. "rms" is the visit-weighted rms of the dB ratio on the sampled pixels; "peak ratio" is the ratio of the two maps' maxima over the whole sphere.

	`empirical_raster_v0016`, $\lambda = 10^{-3}$		`empirical_raster_v0017`, $\lambda = 10^{-2}$	
MHz	rms [dB]	peak ratio	rms [dB]	peak ratio
66.4	3.8	5.25	0.3	0.98
70.3	2.5	2.29	0.4	0.99
74.2	1.7	1.30	0.4	1.01
78.1	1.2	1.07	0.3	1.00
82.0	0.8	1.01	0.3	0.99
85.9	0.5	0.97	0.3	0.98
89.8	0.5	0.99	0.3	0.97
≥90, worst	0.9	1.13	0.9	1.21

Axle tilt. Small corrections are then fitted on top in nested sets, and the set with the best validation score is kept (Table 3). The azimuth and elevation zeros turn the beam about the boresight and about the elevation axle, which runs along the highline; neither can tip the beam along the highline. The products before this one let the transmitter move instead, and its east and north offsets ran to their bound ($1.5/\sqrt{2}$ m per axis), along the highline. Here the transmitter stays where memo 002 puts it, and the sixth correction is a tilt of the elevation axle out of the horizontal, a rotation about the horizontal axis perpendicular to the highline (the mount becomes $R_z(\psi)\,R_y(t)\,R_x(\mathrm{el})\,R_z(\mathrm{az})$; positive t lowers the ψ end; `eigsep_data.beam_mapping.tx_fit`). Without the tilt the best validation score is 0.093; with it, 0.076. The kept set is `az_el_tilt`: $\delta\mathrm{az} = -0.07°$, so the table's zero lies $+15.93°$ from the highline, $\delta\mathrm{el} = +0.05°$, and $t = +4.74°$ (condition number 1.5).

The IMU's tilt. Box-air's elevation IMU rides on the box. Its accelerometer, put into the box frame by the calibration in each file's header, measures the specific force; its component along the box's x axis, the elevation axle, is $-\sin t$. Over the raster's 6990 integrations its median is $t = +5.97°$, and the box swings between +4.7 and +7.3° (5–95%). The fit's single +4.74° has the same sign and lies at the low end of that range. The difference from the median, about a degree, is smaller than the IMU's mounting correction: its calibration rotates the sensor by 3.7° into the box frame. Held at the IMU's value, with the az and el zeros free, the fit scores 0.080 on validation against 0.076 free (Table 4). The fitted tilt is an empirical correction: it can also stand in for a lean of the dipoles in the box, or for an error in the transmitter's height, and it is not a measurement of the box.

Table 3: Nested geometry corrections, fitted with the HFSS beam on training samples and scored by the median per-tooth fractional rms on the validation and test stripes; the transmitter is held at memo 002's position. "cond." is the condition number of the fit's Jacobian.

free	$\delta\alpha$ [deg]	δaz [deg]	δel [deg]	tilt [deg]	cond.	val. / test median rms
none	+0.00	+0.00	+0.00	+0.00	1	0.091 / 0.090
polarization	+0.48	+0.00	+0.00	+0.00	1	0.091 / 0.088
az, el zero	+0.00	-0.09	-0.34	+0.00	1	0.093 / 0.088
polarization, az, el	+0.50	-0.11	-0.35	+0.00	1	0.093 / 0.087
axle tilt	+0.00	+0.00	+0.00	+4.73	1	0.077 / 0.077
az, el zero, tilt	+0.00	-0.07	+0.05	+4.74	2	0.076 / 0.077
polarization, az, el, tilt	+0.17	-0.07	+0.04	+4.72	2	0.077 / 0.075

Handedness. Memo 003 shows that the azimuth is left-handed about the boresight, from `imu_az`'s turn with the shaft, and that the arm ratio alone cannot tell the handedness. The beam pattern breaks the mirror symmetry only through the tilt and the beam's own asymmetries, so this fit is not an independent test. Rerunning

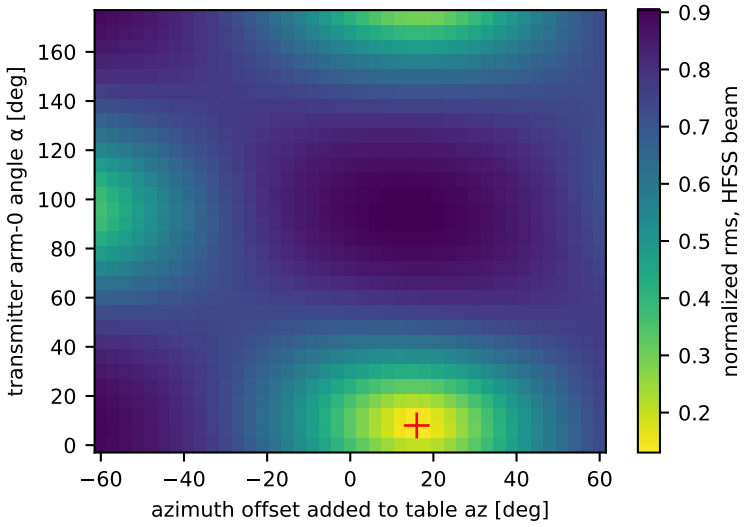


Figure 2: The HFSS beam's fit to the training samples over the azimuth offset and the transmitter arm-0 angle α, with no corrections (so no axle tilt); the cross marks the minimum.

the geometry stage on this product's data with the table's azimuth negated (right-handed; Table 4) gives the mirrored zero (−16°) and arm 0 at 84°. With the tilt free it wants −6.16°, the opposite sign to the IMU, and scores 0.077 on validation against 0.076. With the tilt held at the IMU's value it scores 0.149 against 0.080. The left-handed azimuth is the one whose tilt agrees with the IMU. Memo 003 finds the same zero from the arm ratio. Arm 0's bearing, 172°, lies about 10° from the axis photographed as rx6 (memo 003 leaves that gap open), far nearer it than the other arm, so memo 001's arm labels stand.

Table 4: The geometry stage with each azimuth handedness, with the axle tilt free (all seven candidate sets of Table 3) and with it held at the IMU's +5.97° (az and el zeros free): the coarse scan's best azimuth zero, the bearing of transmitter arm 0 and its rms, the kept corrections and the resulting tilt, and the median per-tooth fractional rms on the validation and test stripes. The right-handed rows negate the table's azimuth, so their validation and test stripes are other samples.

azimuth handedness	axle tilt	az zero [deg]	arm 0 [deg]	coarse rms	corrections kept	tilt [deg]	val. / test
left-hand	free	+16	172	0.120	az, el zero, tilt	+4.74	0.076 / 0.077
left-hand	IMU	+16	172	0.101	az, el zero	+5.97	0.080 / 0.079
right-hand	free	-16	84	0.132	pol., az, el, tilt	-6.16	0.077 / 0.075
right-hand	IMU	-16	85	0.195	az, el zero	+5.97	0.149 / 0.144

Change from v0013. The previous reference product, `empirical_raster_v0013`, used pointing table v2.0, whose azimuth is the right-handed mirror of v5's, geometry release `v0004`'s positions, the transmitter offsets as the free correction, and DPSS and PCA penalties of 0.01 and 0.001. Its coarse scan found the mirror of this one (offset −16°, $\alpha = 96°$), and its transmitter offsets sat at the bound, (+1.06, −1.06) m. The transmitter's direction from the antenna differs from `v0004`'s by 1.55°, and by 2.36° from v0013's fitted position. On the test stripes v0013 scored 0.111 (HFSS), 0.083 (DPSS) and 0.083 (PCA), and its DPSS beam beat HFSS on 59 teeth; this product scores 0.091, 0.076 and 0.078, better on 44. The stripes are bins of table azimuth, which v5 negates, so the two products' test samples differ, and this product also drops the rows outside the ∼91m era. The scores show that the new geometry fits at least as well; the changes are not attributable to any one of them.

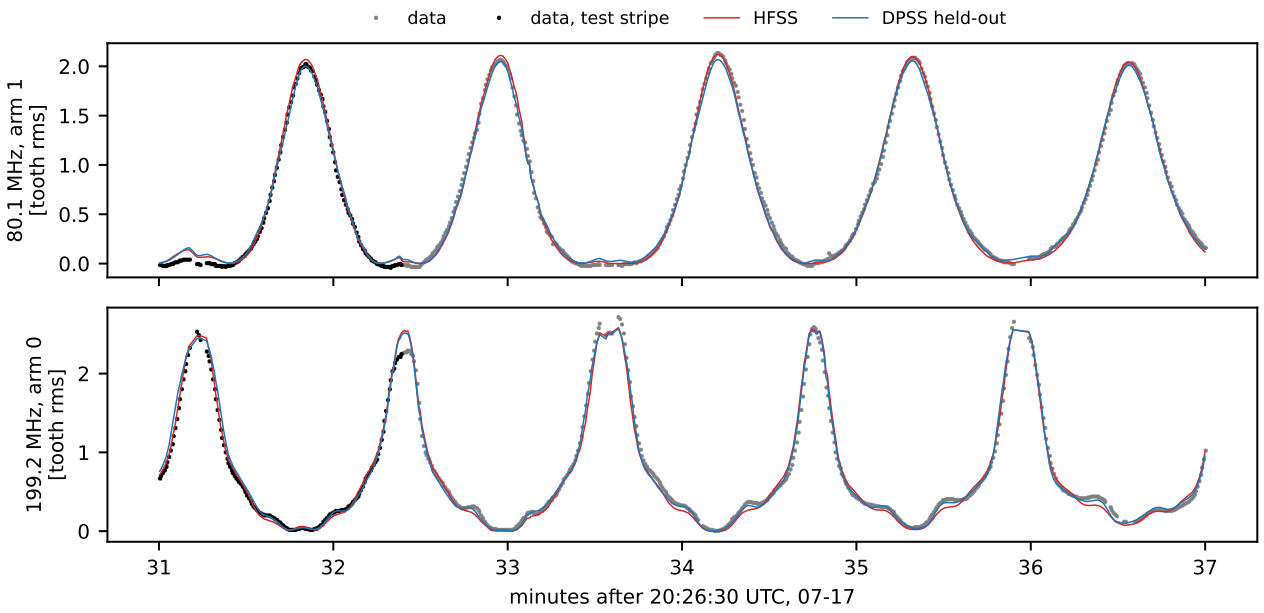


Figure 3: Two teeth over six minutes of the raster, in units of each tooth's rms: data (grey; black on a test stripe), the HFSS beam with its fitted gain (red), and the held-out DPSS beam (blue). Each peak is one elevation revolution passing the transmitter.

5 Results

Figure 3 shows the data and both models for two teeth. Both follow the large modulation as the boresight passes the transmitter once per revolution; the differences are in the shoulders and the sidelobes.

Raster maps. The direct comparison is the raster itself (Fig. 4): each tooth's background-subtracted power at every good sample, placed at its azimuth and elevation in the product's fitted geometry (table azimuth plus the $+16°$ offset and δaz, IMU elevation plus δel; the axle tilt enters the model but not these coordinates), beside the HFSS prediction at the same samples with that tooth's gain fitted on the training stripes, and their difference. Appendix B shows all 101 candidate teeth, 50.8–246.1 MHz, one by one, including the 37 the selection dropped (Fig. 1); for those the HFSS gain is fitted the same way and nothing else is. The product holds only the selected teeth, so the excluded ones are rebuilt from the raster with the product's own generator (`fit_beam.load_data`: same samples, background and masks), which reproduces the selected teeth to <0.001% rms of each tooth.

Table 5: Per-tooth fractional rms $\|d-m\|/\|d\|$ over the 64 teeth: the mean weighted by good-sample count, the 90th percentile and the worst tooth. Only the held-out fits' test column is independent of every fitted choice; HFSS has one gain per tooth fitted on the training stripes.

model	train	validation	test
HFSS	0.092 / 0.151 / 0.261	0.092 / 0.160 / 0.285	0.091 / 0.157 / 0.270
DPSS held-out	0.070 / 0.098 / 0.164	0.072 / 0.105 / 0.174	0.076 / 0.103 / 0.250
PCA held-out	0.072 / 0.107 / 0.169	0.074 / 0.115 / 0.181	0.078 / 0.113 / 0.250
DPSS refit	0.070 / 0.099 / 0.166	0.070 / 0.104 / 0.160	0.072 / 0.100 / 0.241
DPSS exported	0.070 / 0.099 / 0.166	0.070 / 0.104 / 0.160	0.072 / 0.100 / 0.242

Scores. On the test stripes the held-out DPSS beam reaches 0.076 (90th percentile 0.103, worst 0.250 at 148.4 MHz) against HFSS's 0.091 (0.157, 0.270 at 230.5 MHz) (Table 5). PCA does the same (0.078). The empirical beam is better on 44 of 64 teeth (Fig. 5). The worst empirical tooth, 148.4 MHz, is one of the weak arm-0 teeth, whose background matters most.

Table 6: Test-stripe fractional rms by band (mean over teeth weighted by sample count), and the white-noise floor.

band [MHz]	teeth	HFSS	DPSS held-out	PCA held-out	noise floor
50–87	6	0.081	0.065	0.074	0.013
87–145	11	0.084	0.081	0.084	0.015
145–190	23	0.077	0.072	0.073	0.013
190–235	22	0.112	0.080	0.082	0.015
235–245	2	0.107	0.070	0.066	0.013

The residual is not noise. The white-noise part of each tooth's residual, estimated from differences of successive samples, is 0.014 of the tooth's rms on average (worst 0.036, the weak 132.8 MHz arm-0 tooth). The test residual is $\sim$5$\times$ larger, so it is structure that changes slowly along the raster: beam features the model does not capture, background errors, or geometry. Binned by the transmitter's angle from the boresight (Fig. 6), the fractional residual is smallest near the boresight and near the back of the antenna, where most of the tooth power is, and largest at 60–140°, where the power is one to two orders of magnitude lower. The empirical beam improves on HFSS at every angle.

The beam. Figure 7 compares the exported DPSS beam with HFSS at three frequencies. Weighting each sampled pixel by its number of visits and by the beam's power there, the two differ by 0.1–0.3 dB rms, and at least 99% of that weight is within 1 dB; single low-power pixels differ by up to 3 dB. The raster samples 42% of the HEALPix pixels; elsewhere the beam is set by the HFSS prior and the smoothness of the basis, and its differences from HFSS there are not measurements.

6 Caveats

- **Phase and circular polarization** are not constrained: the data are powers. The coefficients are regularized toward HFSS, which fixes what the data cannot.
- **The test is a held-out stripe of azimuth, not a fresh experiment.** The stripes share the session, the background model and the geometry with the training data.
- **Penalty**: the DPSS penalty is chosen by the exported beam's agreement with HFSS in total power (Table 2), not by a held-out score, which measures only the power in the transmitter's polarization along the raster's track. 10^{-2} is the strongest penalty tried, so the beam where the data are silent is mostly HFSS's. Memo 008's low-band numbers use the exported beam.
- **Background**: the transmitter-off control test of the background (about 1% bias, 2–3% scatter) was made with the earlier 15-mode background and has not been rerun for the 21-mode background used since v0012; it matters most for the weak arm-0 teeth. The raster flags were built with an earlier beam (v0006) and the pointing v2.0 selection as reference.
- **Geometry**: the axle tilt is fitted, with the transmitter held at memo 002's position. It agrees in sign with the IMU and differs by about a degree in size, and it can absorb a lean of the dipoles or an error in the transmitter's height. The fitted elevation zero (+0.05°) does not match the IMU offset implied by the LIDAR (memo 003).
- The $\lambda = 10^{-4}$ fits reached the 3000-iteration limit; the chosen fits converged.

7 Open questions

- What the remaining structure is: a higher $\ell_{\max}$ or a richer spectral basis where the residual is largest, against background and geometry errors.
- What the axle tilt is: the box's own tilt, a lean of the dipoles in the box, or a stand-in for a geometry error. The IMU's tilt and the fitted one differ by about a degree.
- The gap of about 10° between the polarization's arm 0 and the photographed rx6 axis (memo 003).

- A transmitter-off control for the 21-mode background.

Revision history

2026-10-31 Moved to beam product v0017: pointing table v5 (azimuth left-handed about the boresight, memo 003), memo 002's `v0005` posterior-mean positions, and a tilt of the elevation axle in place of the transmitter offsets, which had run to their bound. The IMU measures the tilt's sign. The DPSS penalty is held at 10^{-2}, because the weaker penalty the validation scores prefer (v0016) puts spurious lobes in the exported beam at the bottom of the band. The comparison is now with v0013.

2026-10-08 Moved to beam product v0013, the same fit on geometry release `v0004`; adds the comparison with v0012's `v0001` geometry.

2026-10-02 First version, on beam product v0012. Adds count-weighted scores with worst cases, the white-noise floor, the residual by angle, and the handedness check of the geometry stage.

A Reproducing this memo

The beam product is regenerated, from the eigsep checkout root, by `python data-analysis/scripts/marjum-2026-07/fit_beam.py --output marjum-2026-07/derived/beam/empirical_raster_vNNNN`, with `--penalty dpss=1e-2` and otherwise its defaults (geometry release `v0005` with memo 002's posterior-mean positions, the axle-tilt candidates, teeth `tooth_selection_v0002.json`, flags `raster_flags_v0001.npz`, DPSS and PCA bases, 3000 iterations, background concentration 10^{-6}); the commits it ran at are below. The axle tilt is in `eigsep_data`'s main branch. This memo's figures, tables and numbers come from `python make_products.py` in this directory (set `EIGSEP_CAMPAIGN_ROOT` to the campaign directory that holds `derived/beam/` when running from a worktree), then `pdflatex memo-005.tex` twice; set `EIGSEP_DATA_ANALYSIS` to use a data-analysis checkout other than the one in this checkout. The script checks the SHA-256 of every gitignored input.

input	SHA-256 (first 16)
derived/beam/empirical_raster_v0017/provenance.json	ca77de003dfa1206
derived/beam/empirical_raster_v0017/optimization.json	90e378a4d2de27a2
derived/beam/empirical_raster_v0017/metrics.csv	1c27c46e93aa2287
derived/beam/empirical_raster_v0017/geometry_comparison.csv	361bdd455ca1bbdc
derived/beam/empirical_raster_v0017/az_offset_alpha_scan.csv	2cca3715972461f2
derived/beam/empirical_raster_v0017/spatial_order.csv	e0f3502307607449
derived/beam/empirical_raster_v0017/dpss/diagnostics.npz	d3f0dd9e3abc7a38
derived/beam/empirical_raster_v0017/dpss/empirical_beam.npz	025abec4f44be069
derived/beam/empirical_raster_v0017/dpss/basis_selection.csv	ae3ad67ee6086066
derived/beam/empirical_raster_v0017/pca/diagnostics.npz	32192886739b3f39
derived/beam/empirical_raster_v0017/pca/basis_selection.csv	f254c86fe09d1c2a
derived/beam/empirical_raster_v0016/provenance.json	a2956096ed491d50
derived/beam/empirical_raster_v0016/dpss/empirical_beam.npz	57b943bbb8469b8a
derived/beam/empirical_raster_v0013/provenance.json	37b9f6b2432200c3
derived/beam/empirical_raster_v0013/dpss/diagnostics.npz	c829b29129c0ac6c
derived/beam/empirical_raster_v0013/pca/diagnostics.npz	9b428a4f568065ab
derived/beam/tooth_selection_v0002.csv	acd4e893bc6c1899
derived/beam/tooth_selection_v0002.json	08e4d5e3e48ee9e5
eigsep_data/hfss_beam_maps/bowtie_beam.npz	c4046f5e9332cd57
memos/memo-002-marjum-2026-07-geometry/tables/summary.json	7b1f0684903e71c5
marjum-2026-07/imgs/fits/v0004_marjum_geometry/shared.json	8fcf5360cbfa14a0
beam product generated by `data-analysis/scripts/marjum-2026-07/fit_beam.py`	
data-analysis	7be0c37
eigsep_data	b524cc2
eigsep_base	840c10b
eigsep	e1edb4d
eigsep (memo) commit	f1b00d4
eigsep_data commit (HFSS file, geometry model)	632aa7c
data-analysis commit (raster rebuild, `fit_beam.py`)	5548b5f
generated	2026-10-31 14:58 UTC

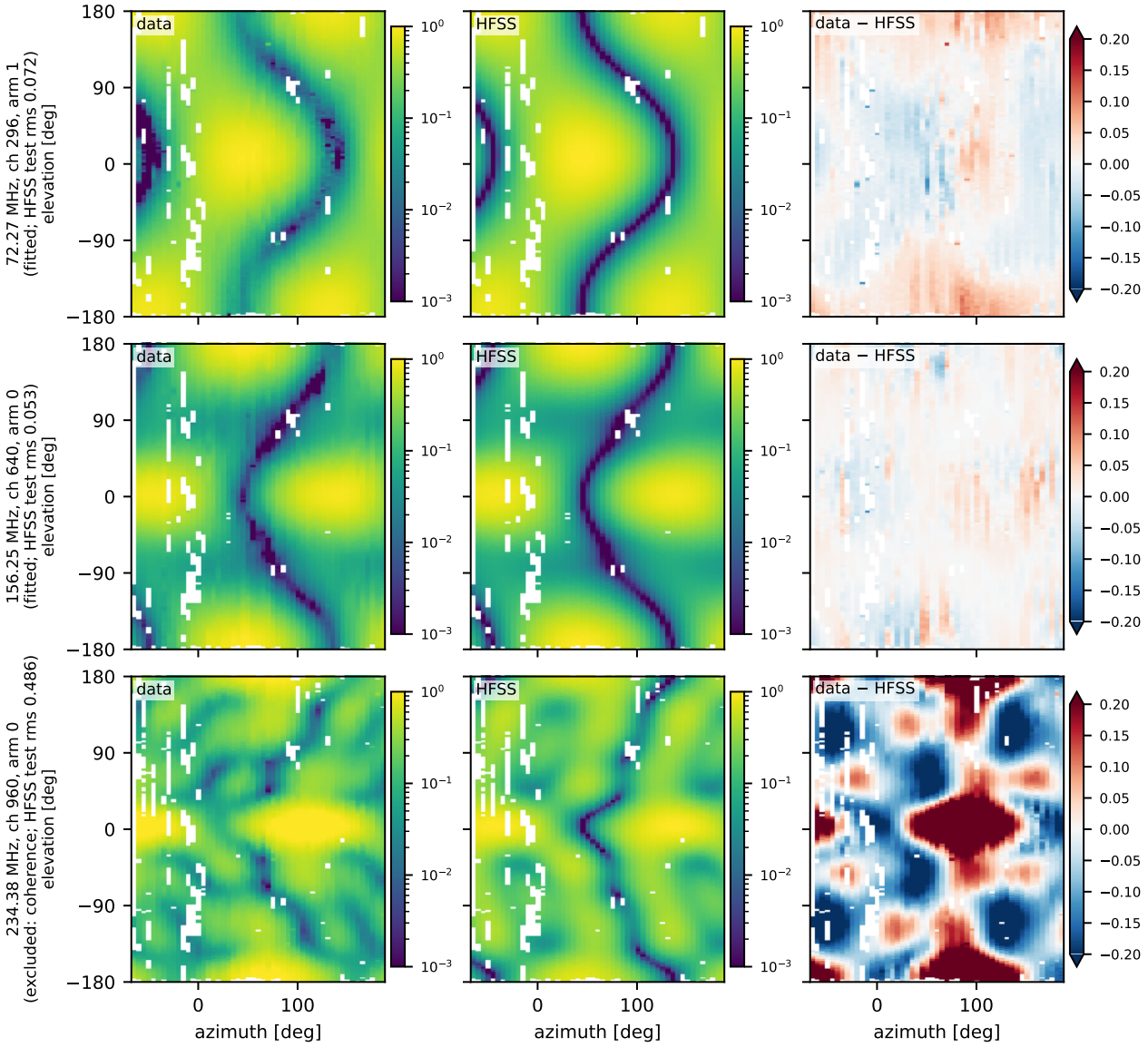


Figure 4: Raster maps of three teeth on azimuth (from the highline, fitted geometry, left-handed about the boresight) and elevation (0 at zenith, ±180° at nadir): data (tooth minus background), the HFSS model with one fitted gain, and data minus model. Every panel is divided by the HFSS model's peak over the raster (its fitted gain times its peak coupling), so all teeth share fixed scales: data and model on a log scale from 10^{-3} to 1 (lower values are drawn at the floor), the residual from −0.2 to +0.2 of the peak. A good fit leaves a pale residual panel; a poor one saturates. Only good samples are drawn. The labels give each tooth's HFSS test-stripe fractional rms; 234.38 MHz was excluded from the fit by the coherence test.

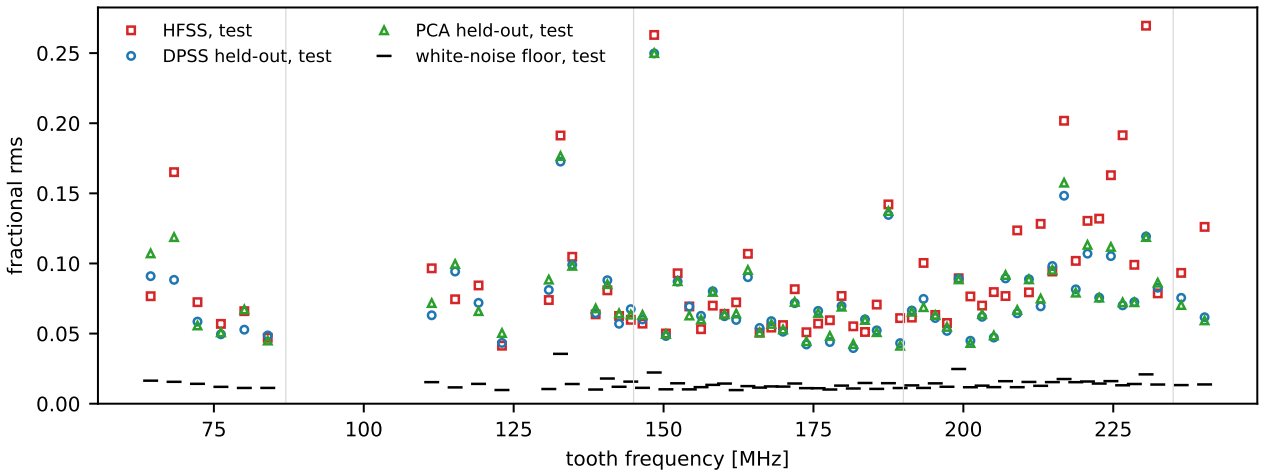


Figure 5: Per-tooth fractional rms on the test stripes for HFSS and the two held-out empirical beams, with the white-noise floor of each tooth's residual. Thin lines mark the band edges of Table 6.

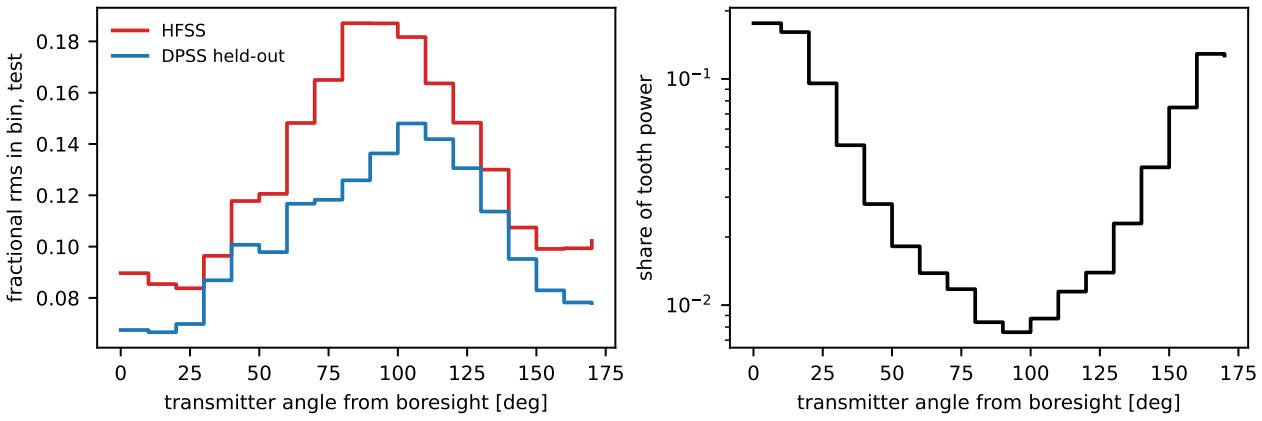


Figure 6: Left: fractional rms on the test stripes, pooled over teeth (each tooth scaled to unit mean power), in 10° bins of the transmitter's angle from the boresight. Right: the share of tooth power in each bin. The antenna has no ground plane, so it receives about as much from behind as from in front.

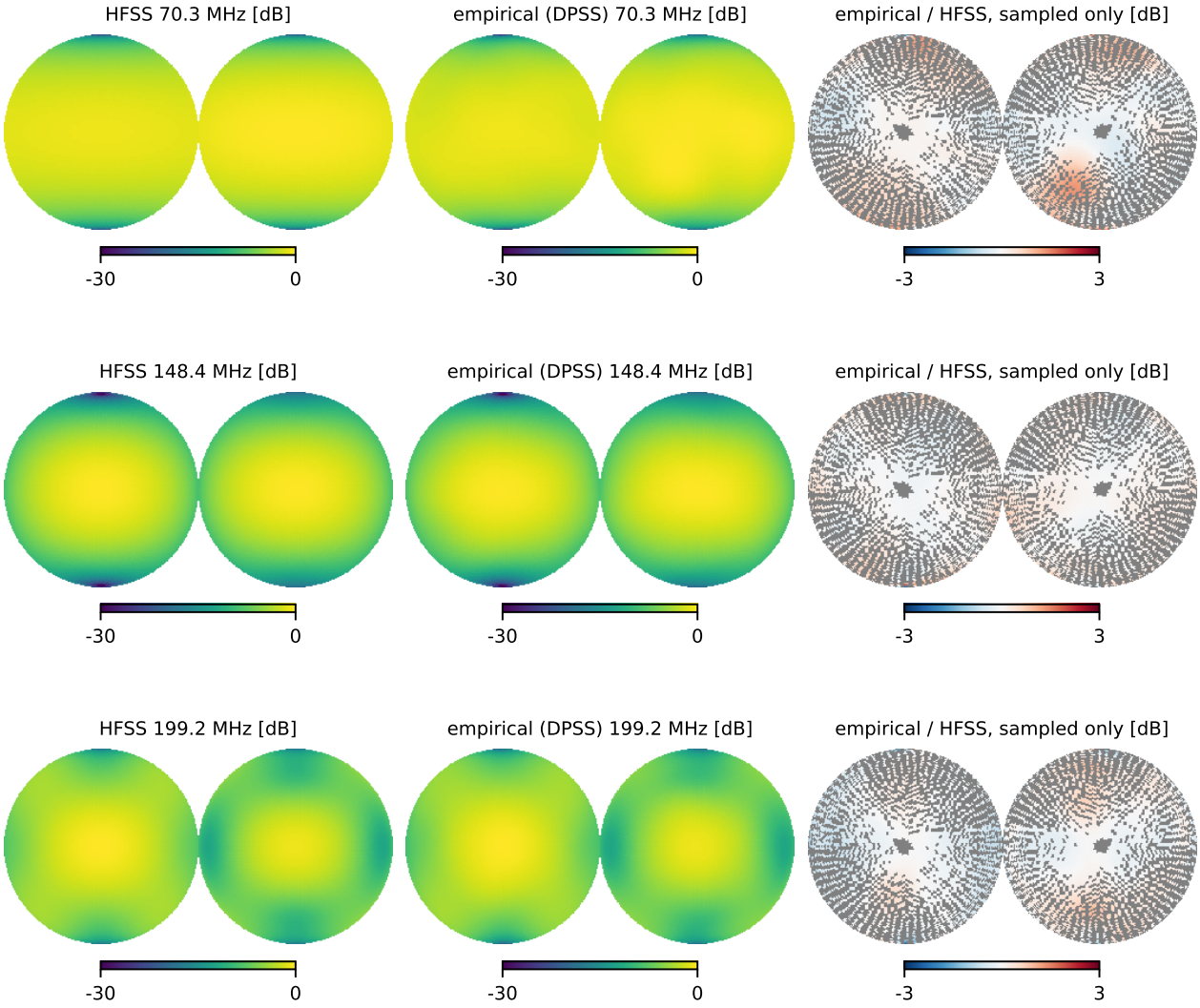


Figure 7: Total power $|\mathbf{E}|^2$ of HFSS (left) and the exported DPSS beam (centre), each peak-normalized, and their ratio on the sampled pixels only (right), at three frequencies. Each panel shows the forward hemisphere (boresight at the centre of the left disc) and the backward hemisphere (right disc).

B Raster maps of every candidate tooth

All 101 candidate teeth in frequency order, 26 pages of four, with the panels, geometry and colour scales of Fig. 4. Each row's label says whether the tooth was fitted or why it was excluded, and gives its HFSS test-stripe fractional rms. Teeth in the FM band (87–108.5 MHz) carry broadcast signals as well as the transmitter, and the background there is fitted without the FM channels. Where the fitted gain is zero (gains are constrained non-negative) there is no HFSS peak to divide by; those rows are divided by the data's 99.5th percentile instead (marked "gain 0"), their HFSS panel sits at the floor, and the test rms is 1. Where the gain is merely small, the scaled data run far off the top of the scale.

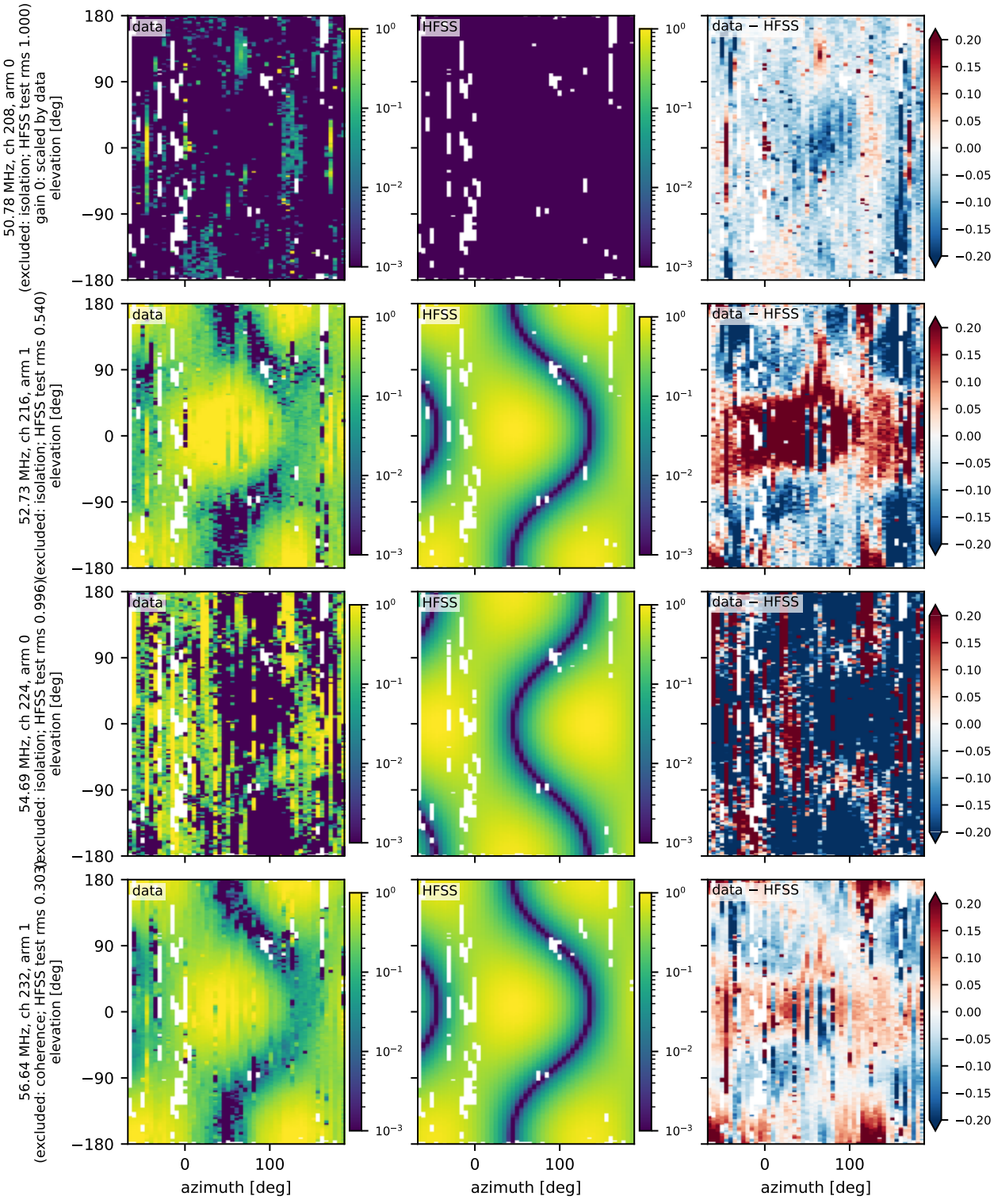


Figure 8: Raster maps, 50.78–56.64 MHz (page 1 of 26); panels as in Fig. 4.

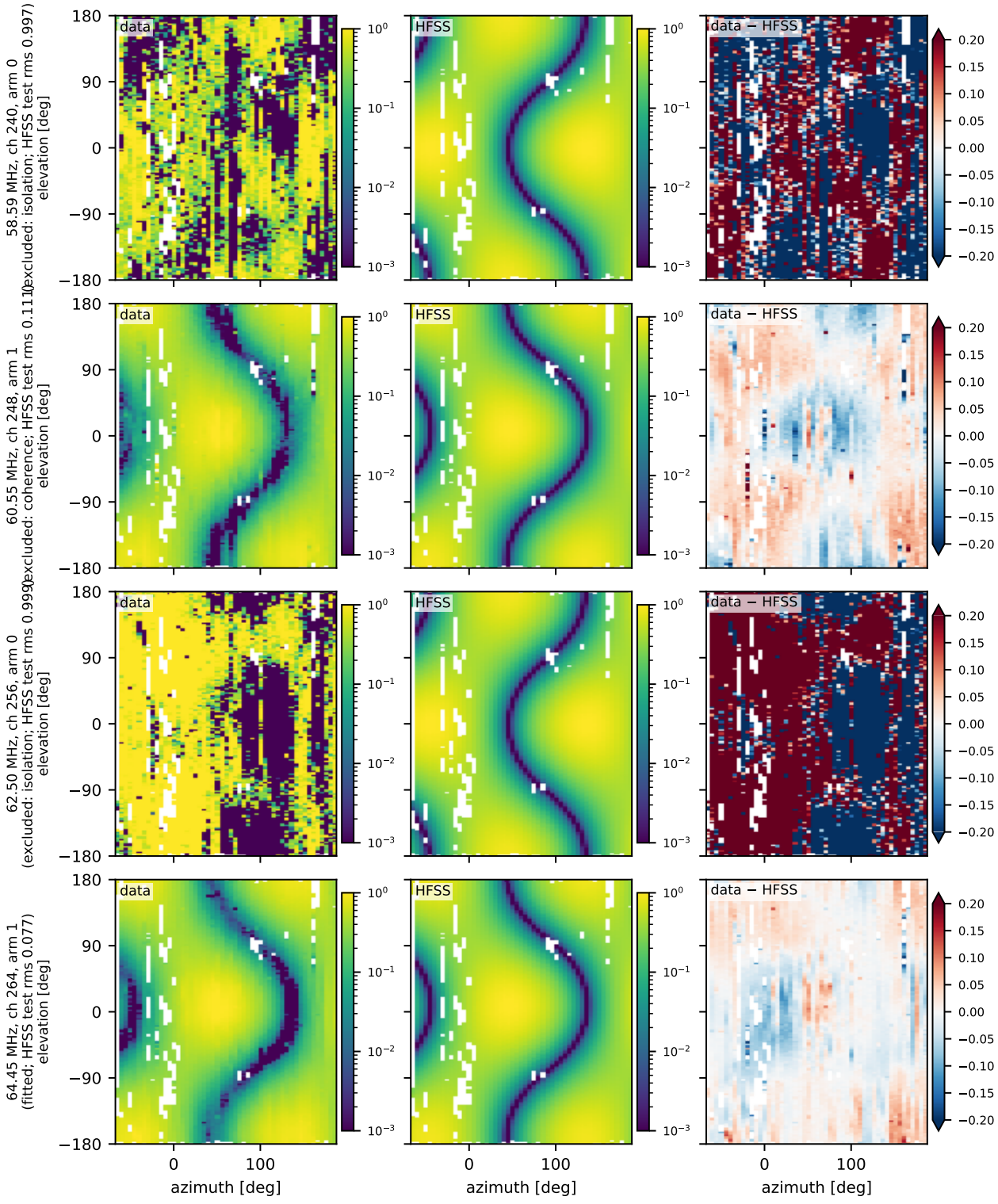


Figure 9: Raster maps, 58.59–64.45 MHz (page 2 of 26); panels as in Fig. 4.

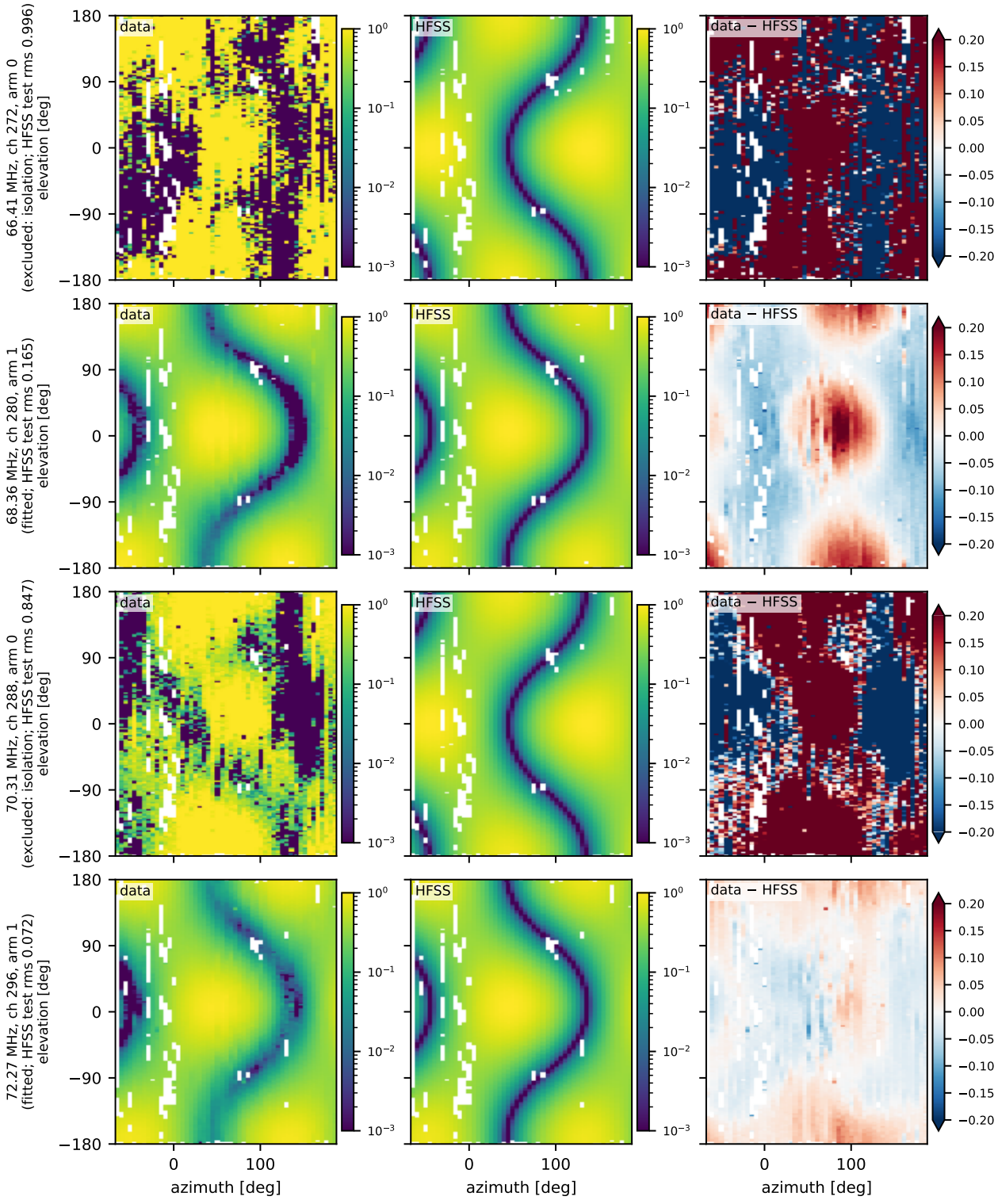


Figure 10: Raster maps, 66.41–72.27 MHz (page 3 of 26); panels as in Fig. 4.

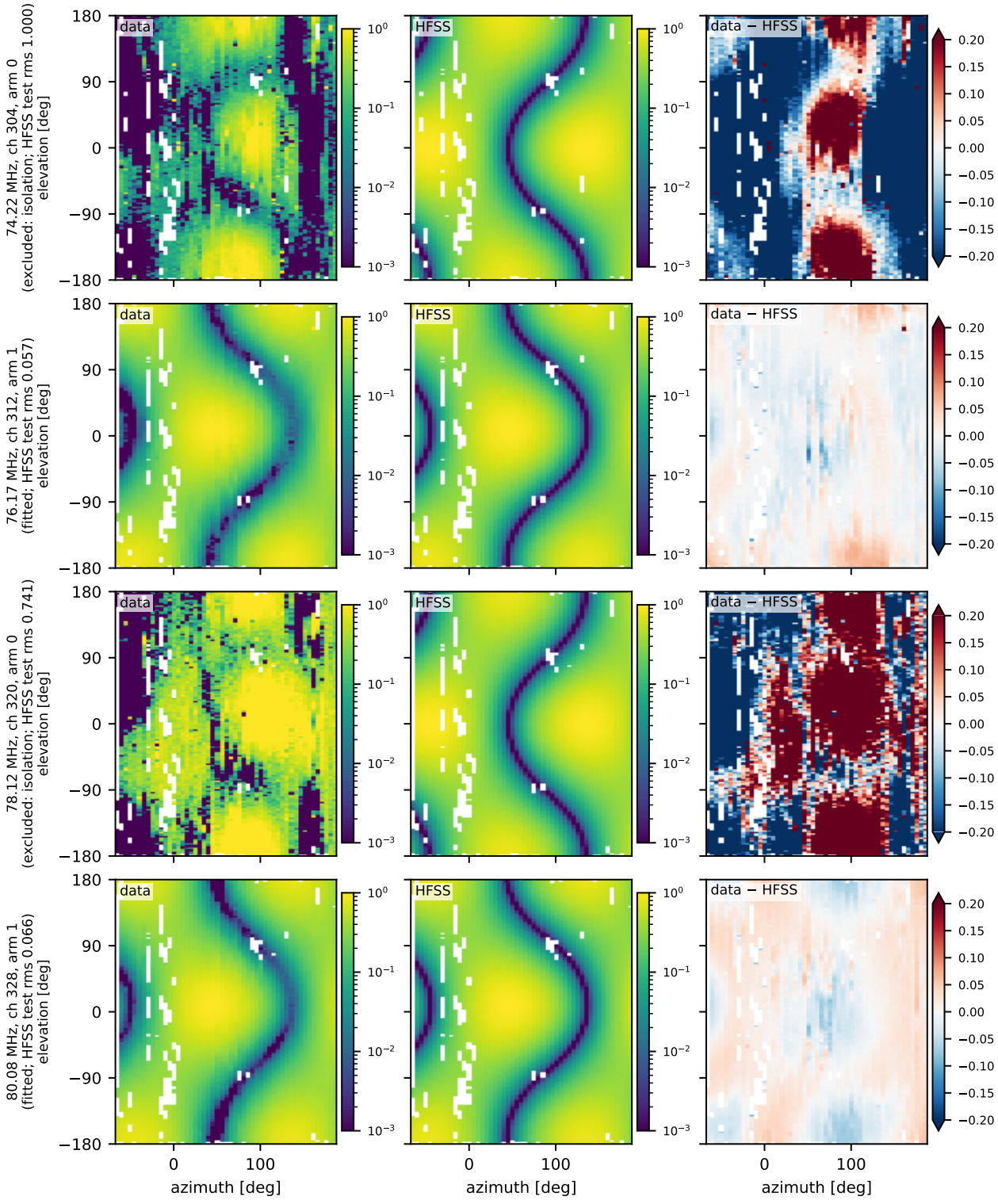


Figure 11: Raster maps, 74.22–80.08 MHz (page 4 of 26); panels as in Fig. 4.

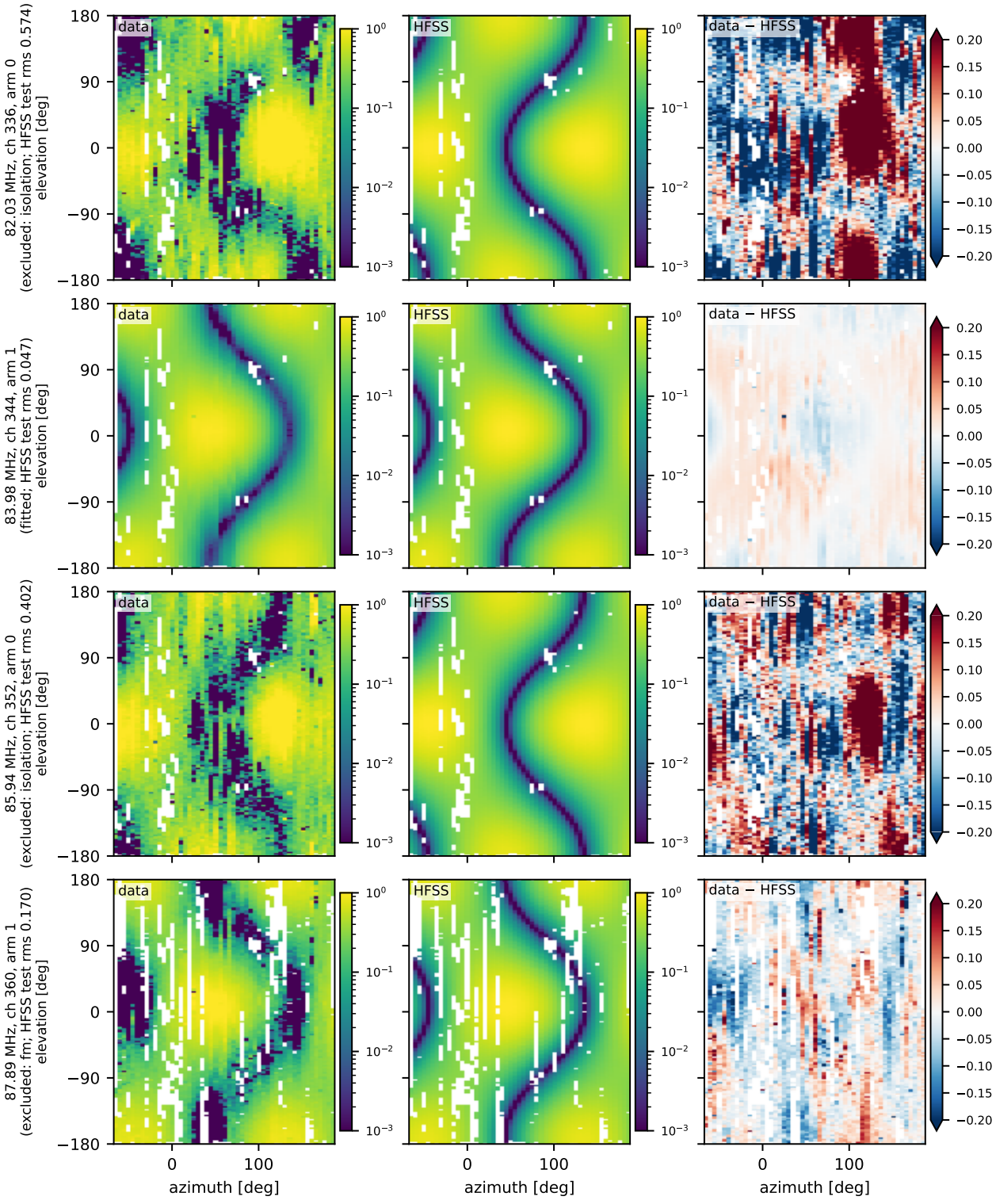


Figure 12: Raster maps, 82.03–87.89 MHz (page 5 of 26); panels as in Fig. 4.

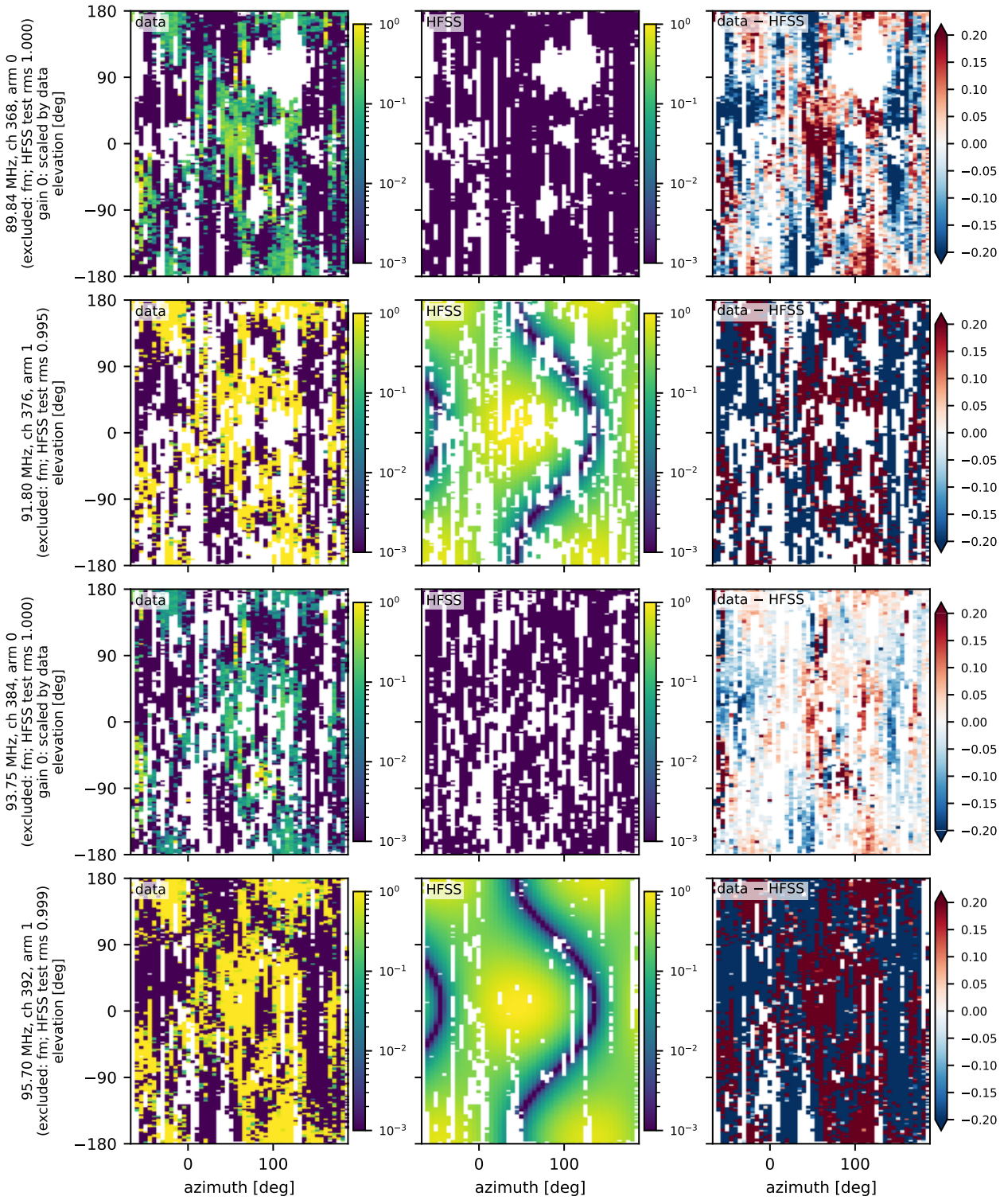


Figure 13: Raster maps, 89.84–95.70 MHz (page 6 of 26); panels as in Fig. 4.

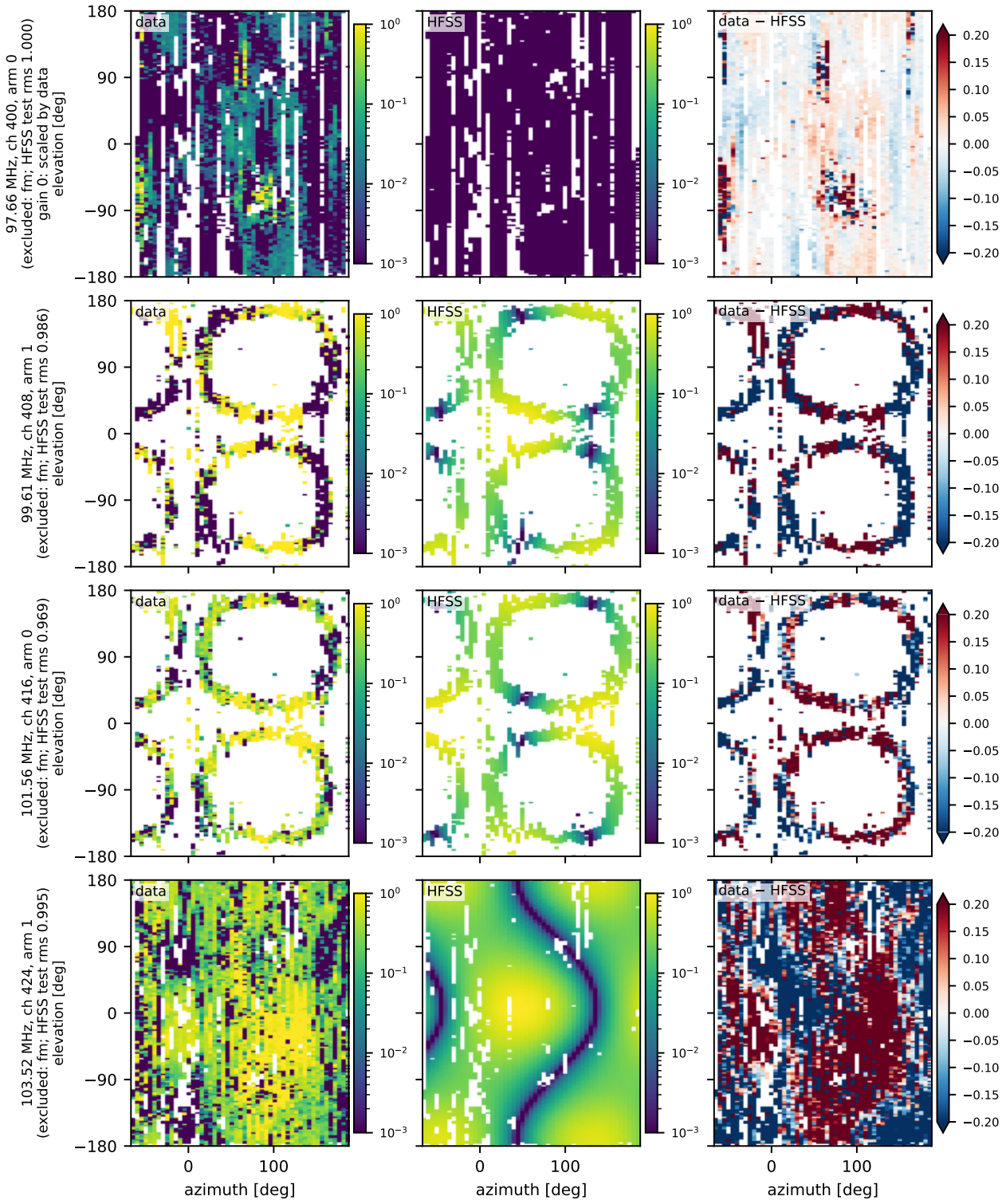


Figure 14: Raster maps, 97.66–103.52 MHz (page 7 of 26); panels as in Fig. 4.

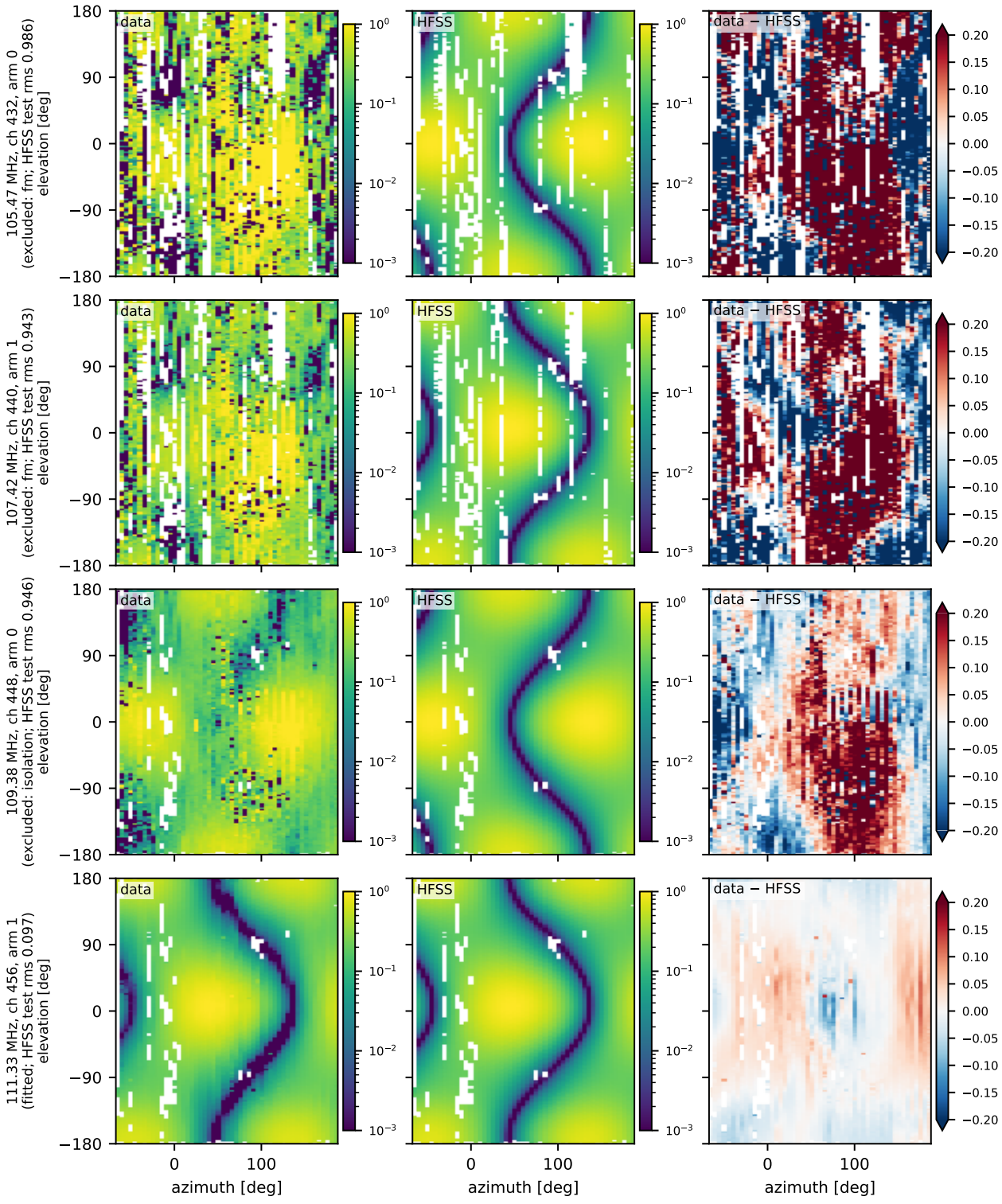


Figure 15: Raster maps, 105.47–111.33 MHz (page 8 of 26); panels as in Fig. 4.

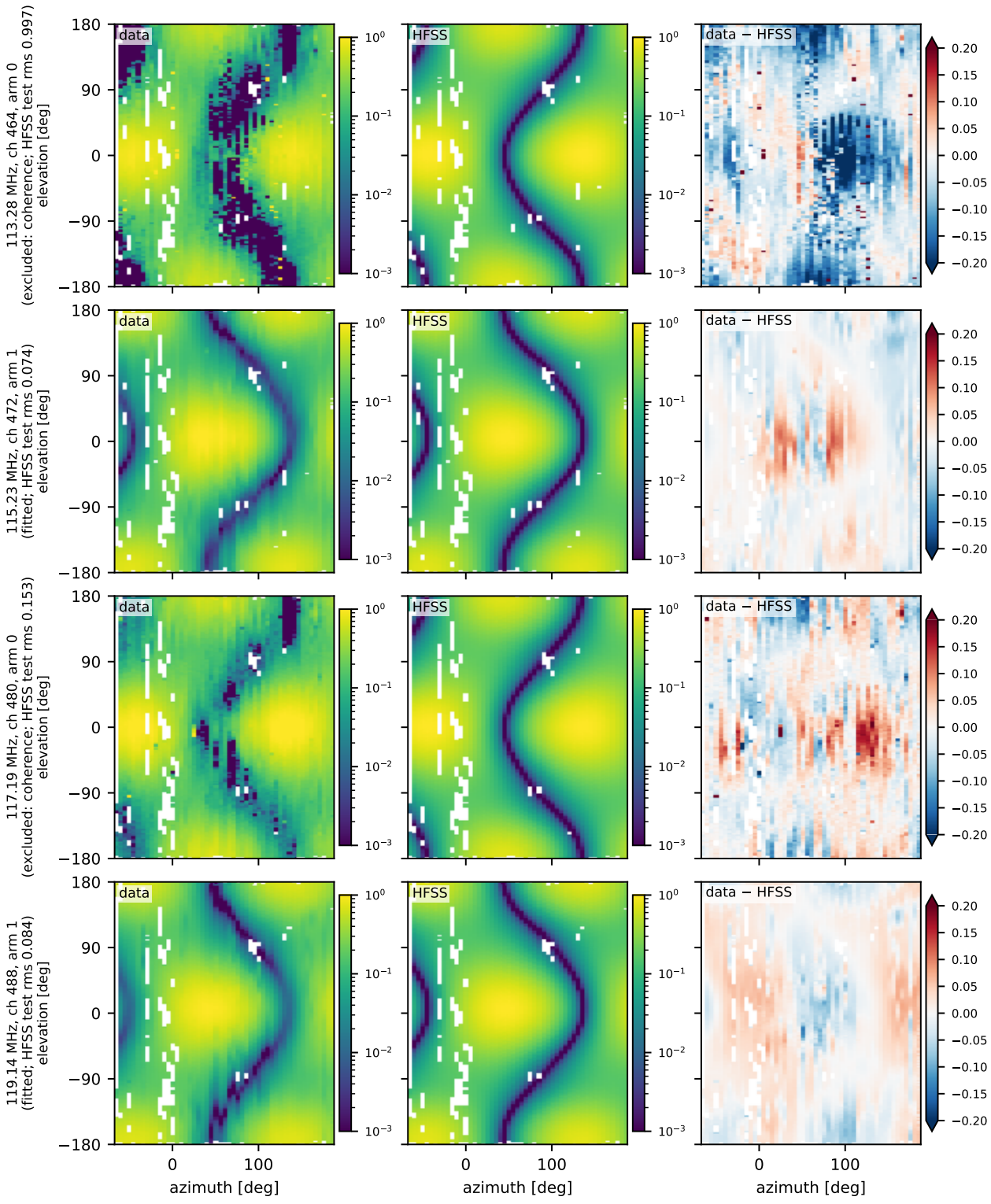


Figure 16: Raster maps, 113.28–119.14 MHz (page 9 of 26); panels as in Fig. 4.

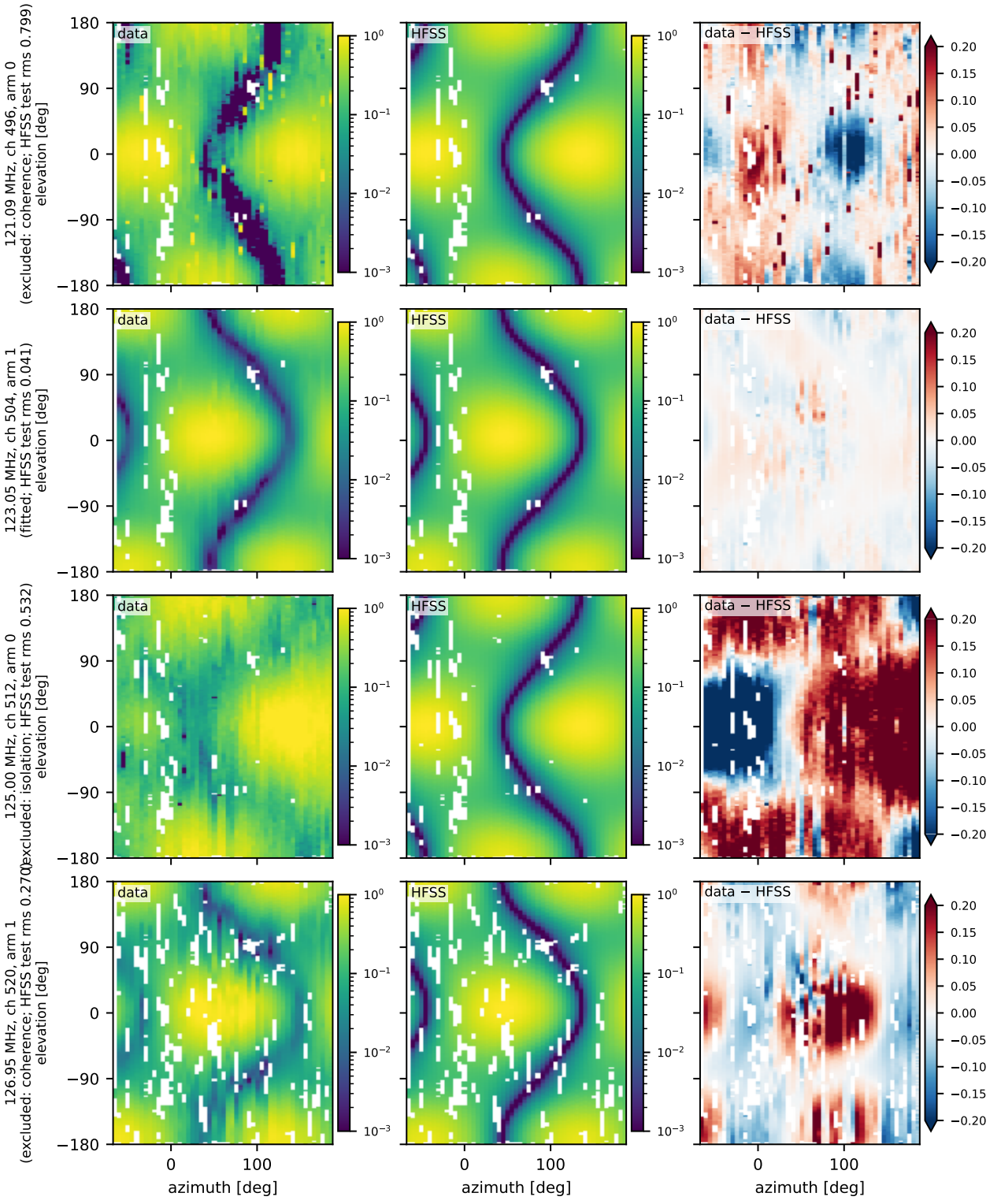


Figure 17: Raster maps, 121.09–126.95 MHz (page 10 of 26); panels as in Fig. 4.

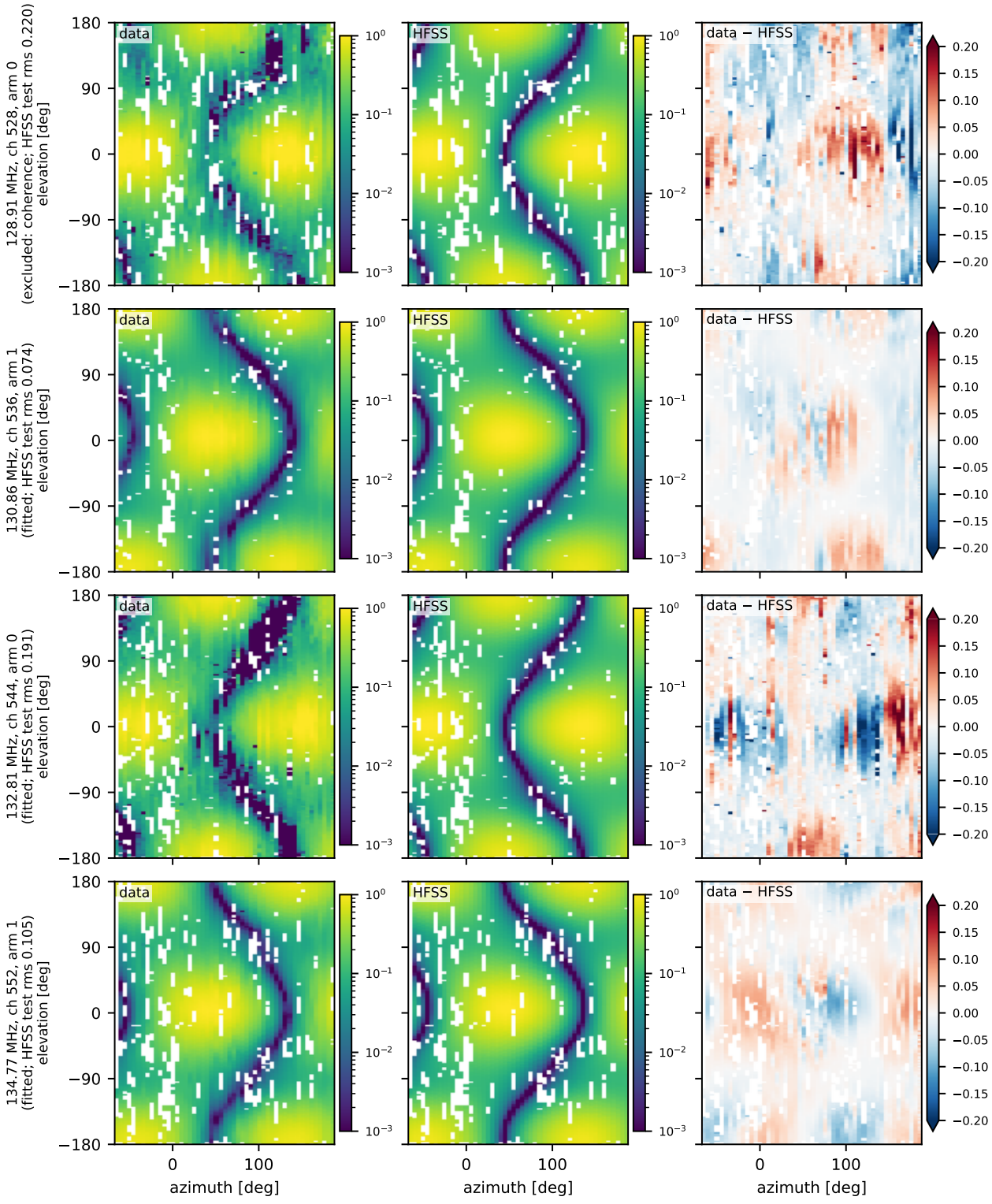


Figure 18: Raster maps, 128.91–134.77 MHz (page 11 of 26); panels as in Fig. 4.

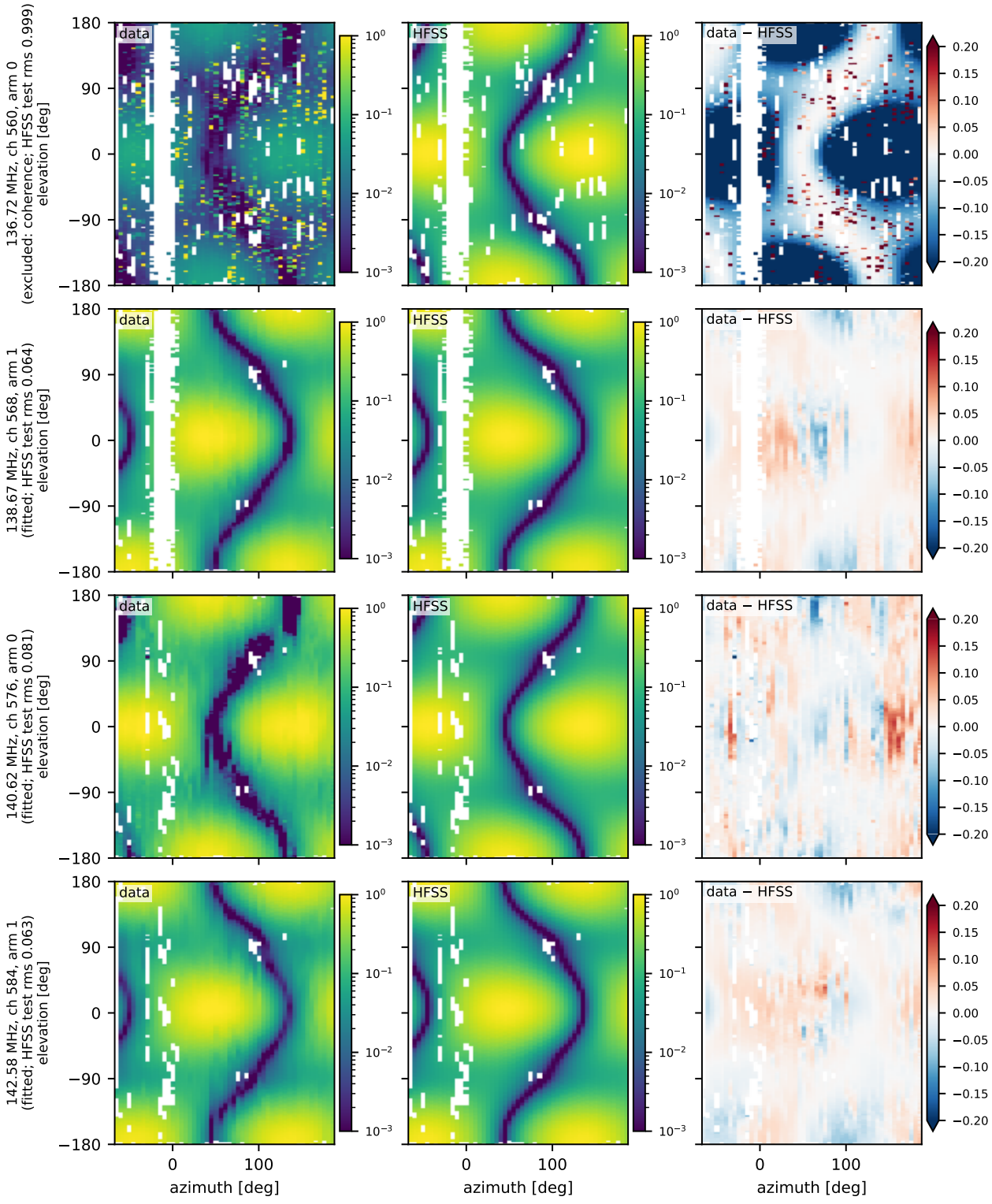


Figure 19: Raster maps, 136.72–142.58 MHz (page 12 of 26); panels as in Fig. 4.

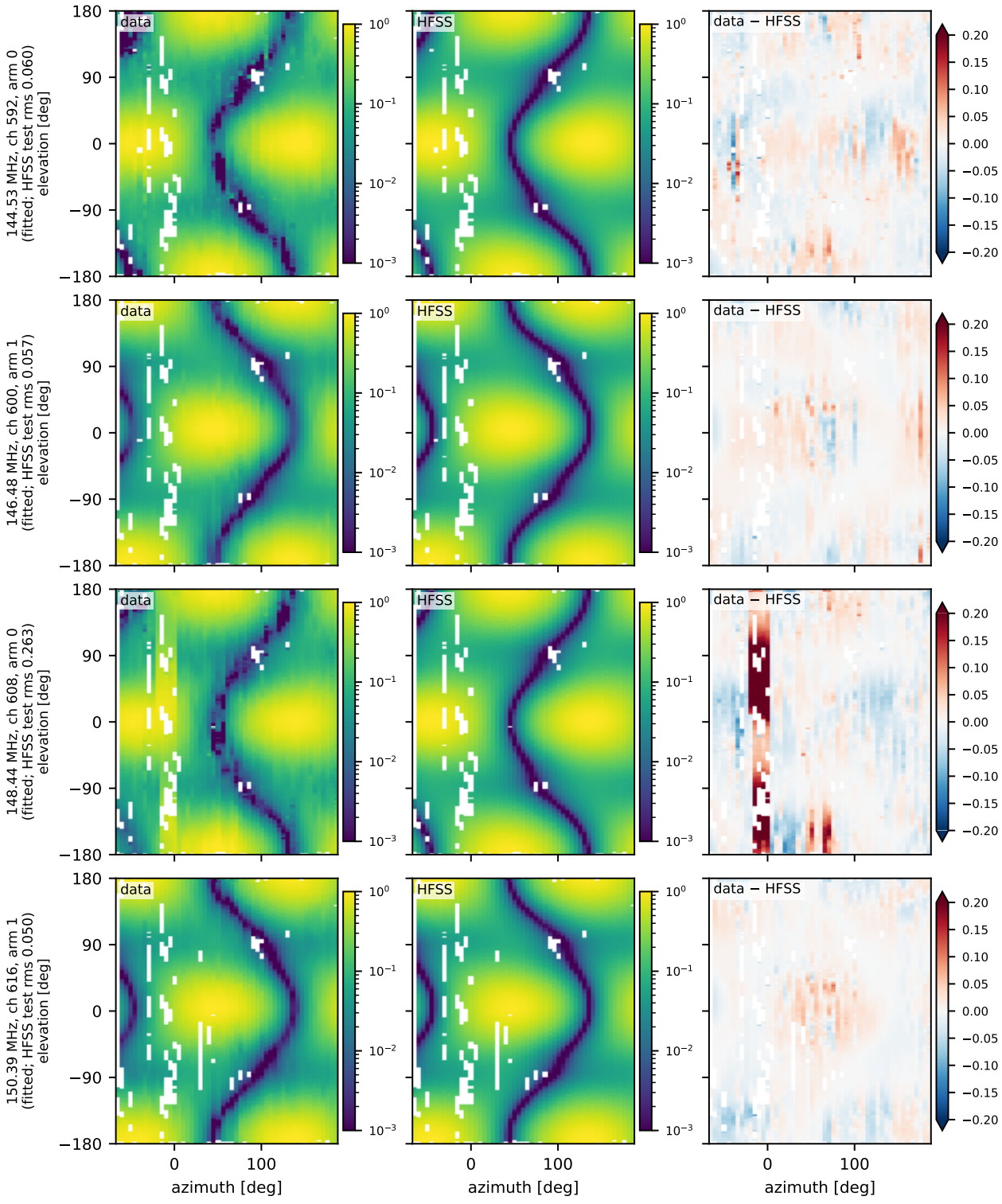


Figure 20: Raster maps, 144.53–150.39 MHz (page 13 of 26); panels as in Fig. 4.

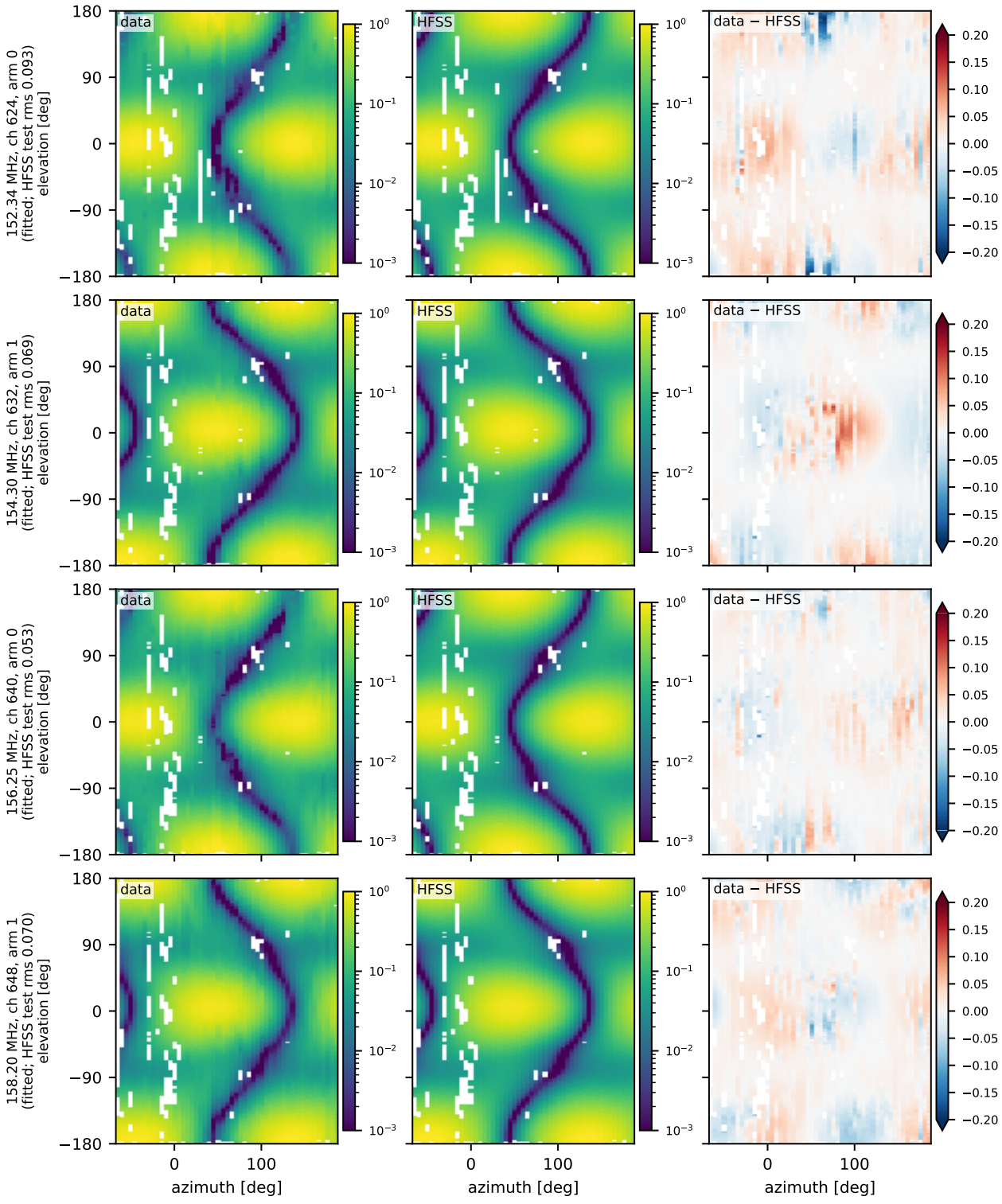


Figure 21: Raster maps, 152.34–158.20 MHz (page 14 of 26); panels as in Fig. 4.

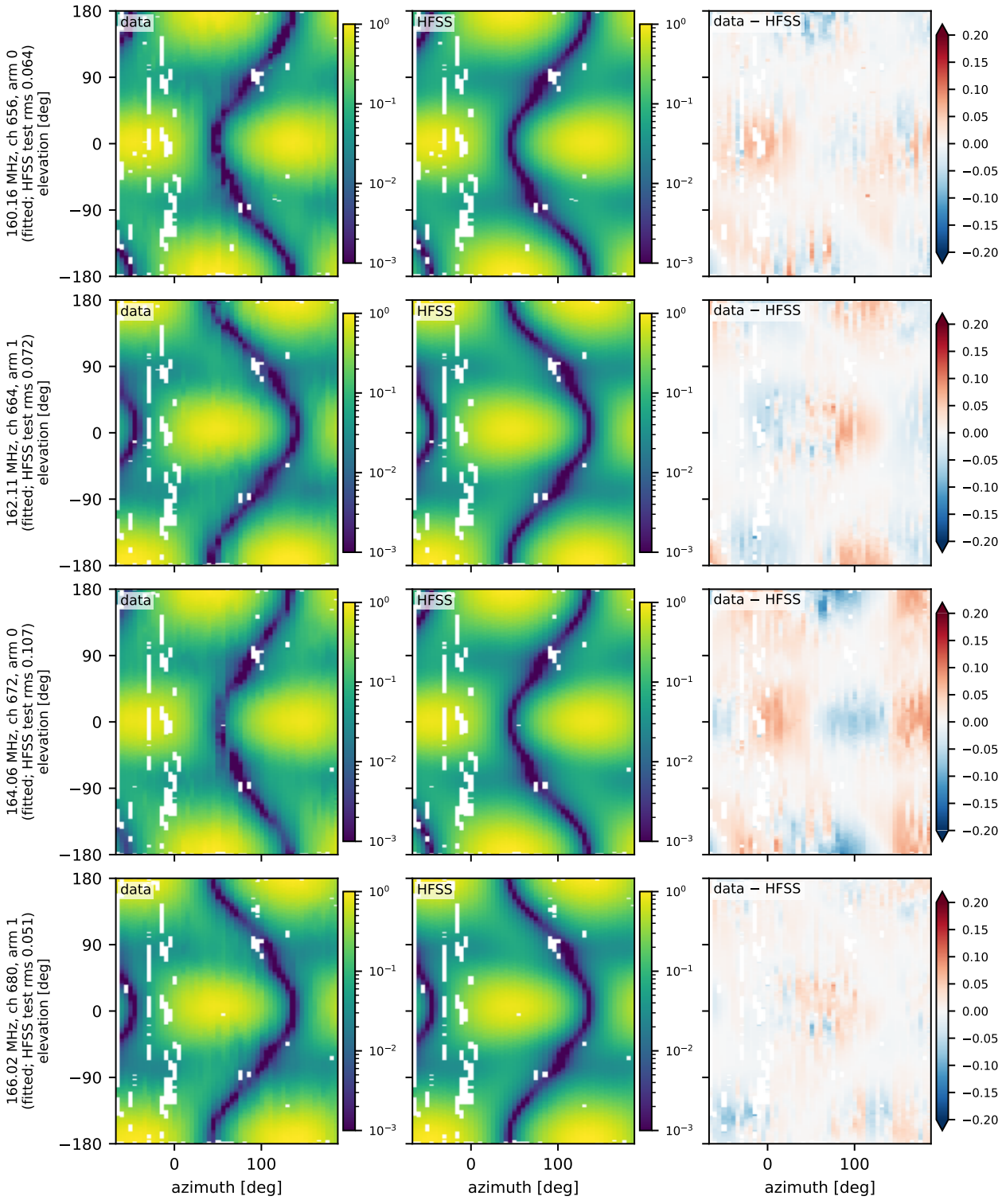


Figure 22: Raster maps, 160.16–166.02 MHz (page 15 of 26); panels as in Fig. 4.

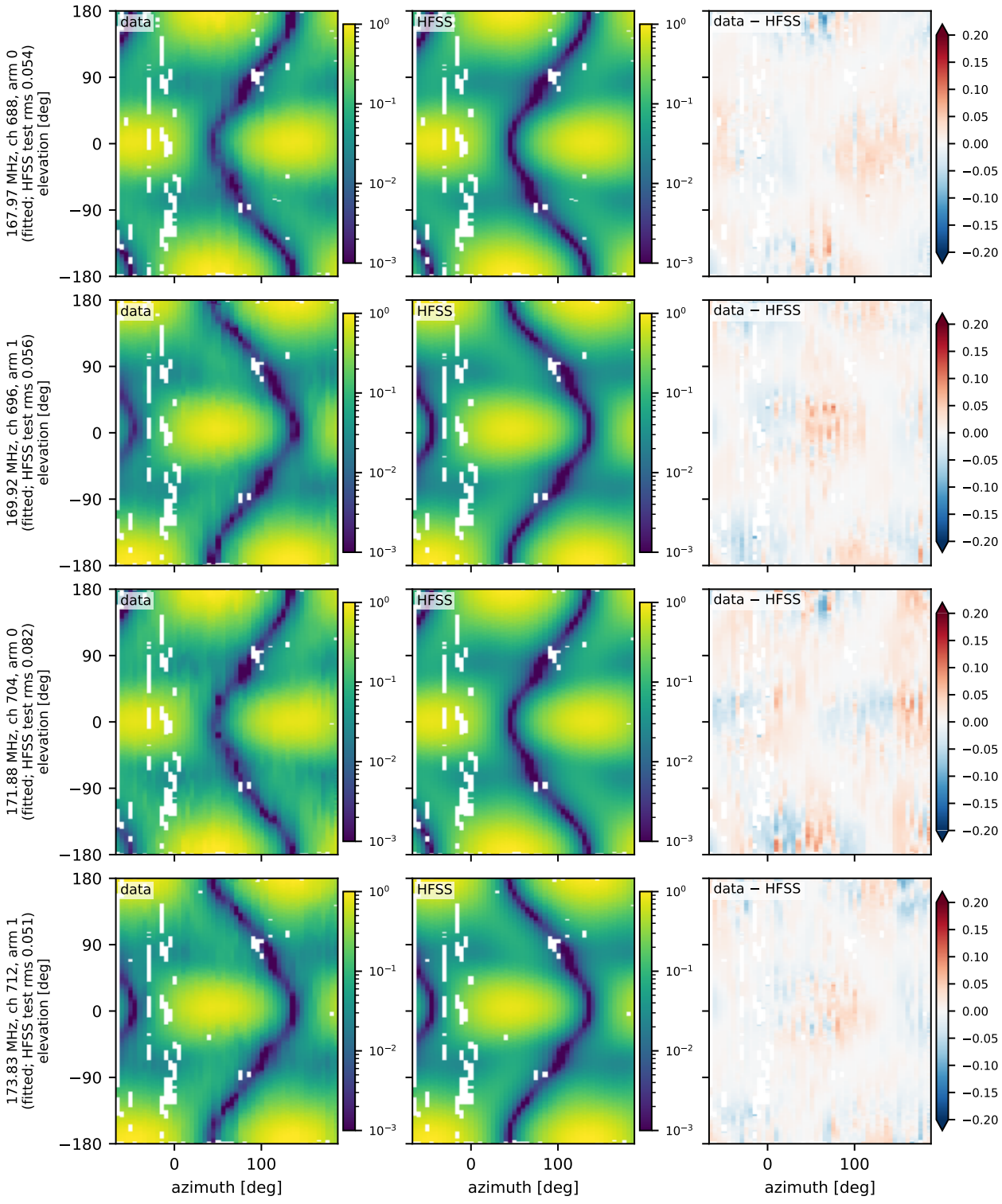


Figure 23: Raster maps, 167.97–173.83 MHz (page 16 of 26); panels as in Fig. 4.

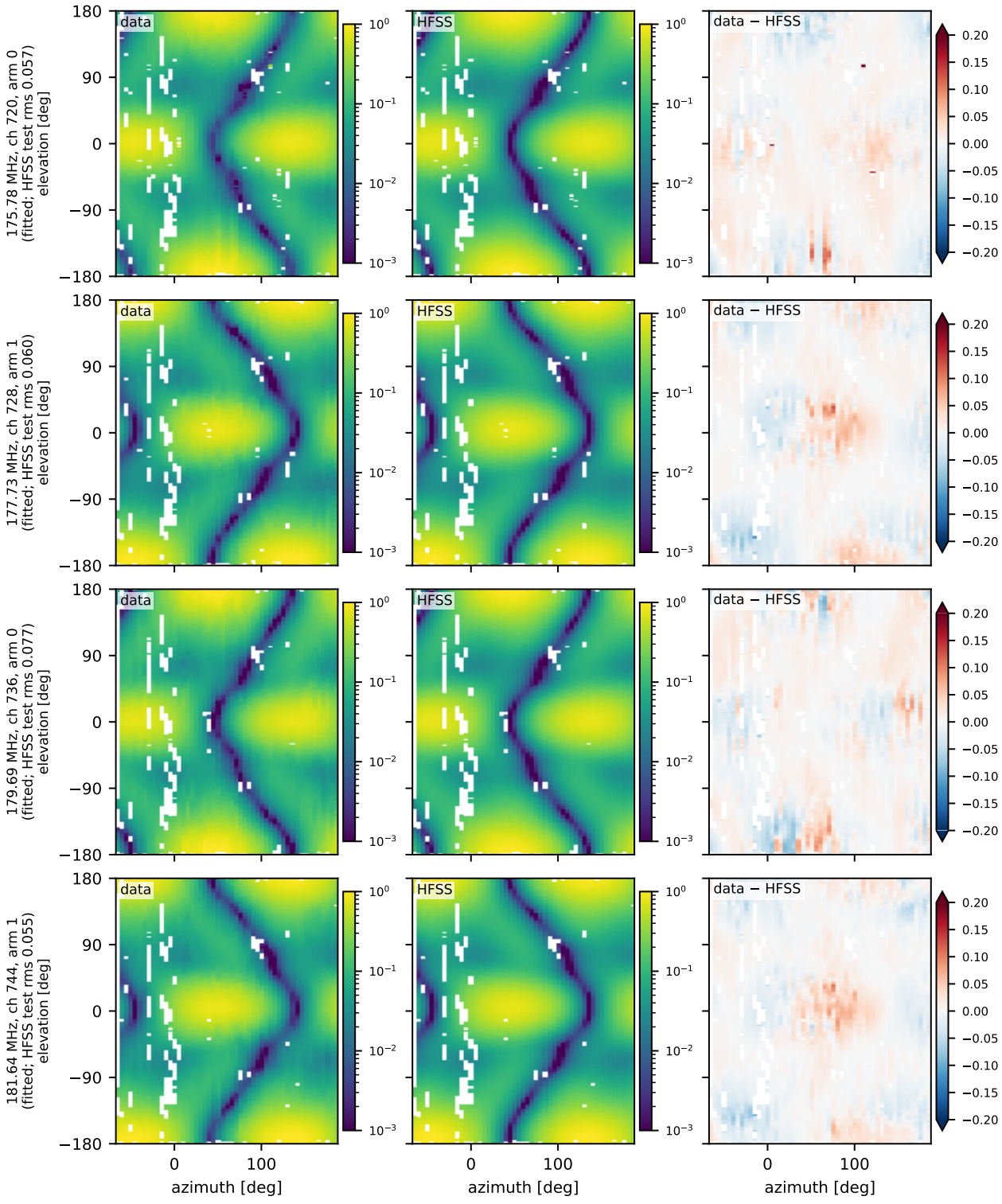


Figure 24: Raster maps, 175.78–181.64 MHz (page 17 of 26); panels as in Fig. 4.

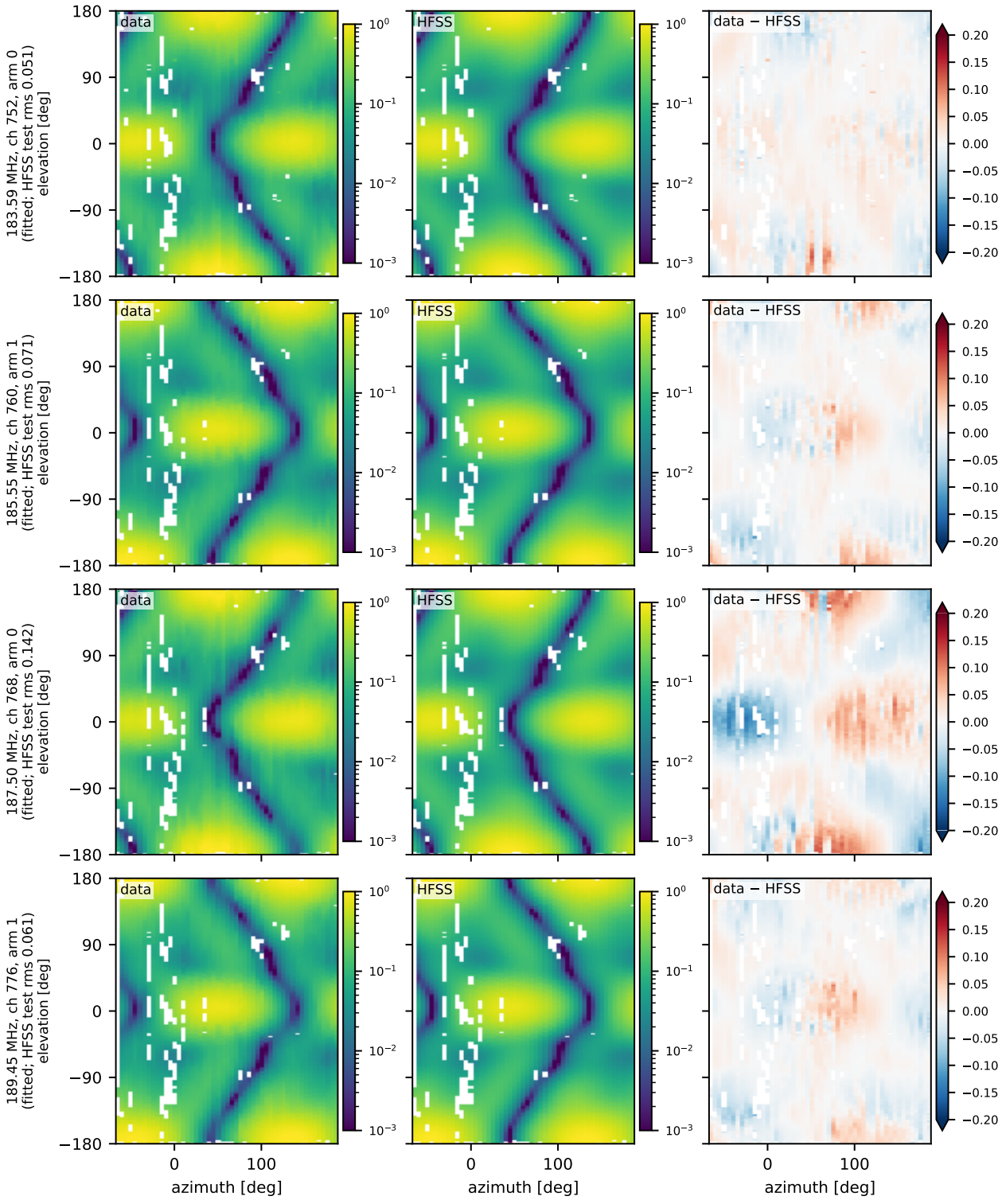


Figure 25: Raster maps, 183.59–189.45 MHz (page 18 of 26); panels as in Fig. 4.

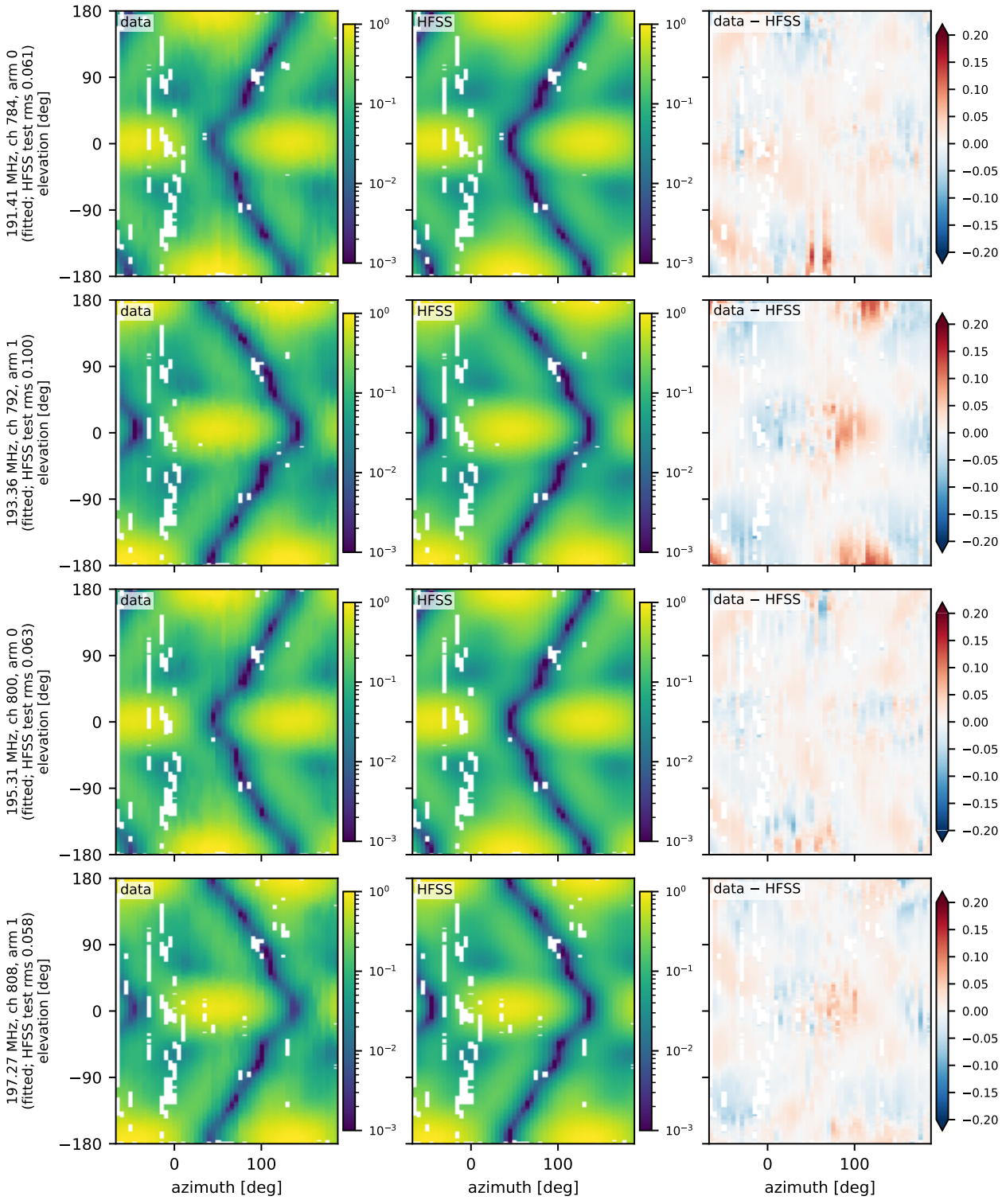


Figure 26: Raster maps, 191.41–197.27 MHz (page 19 of 26); panels as in Fig. 4.

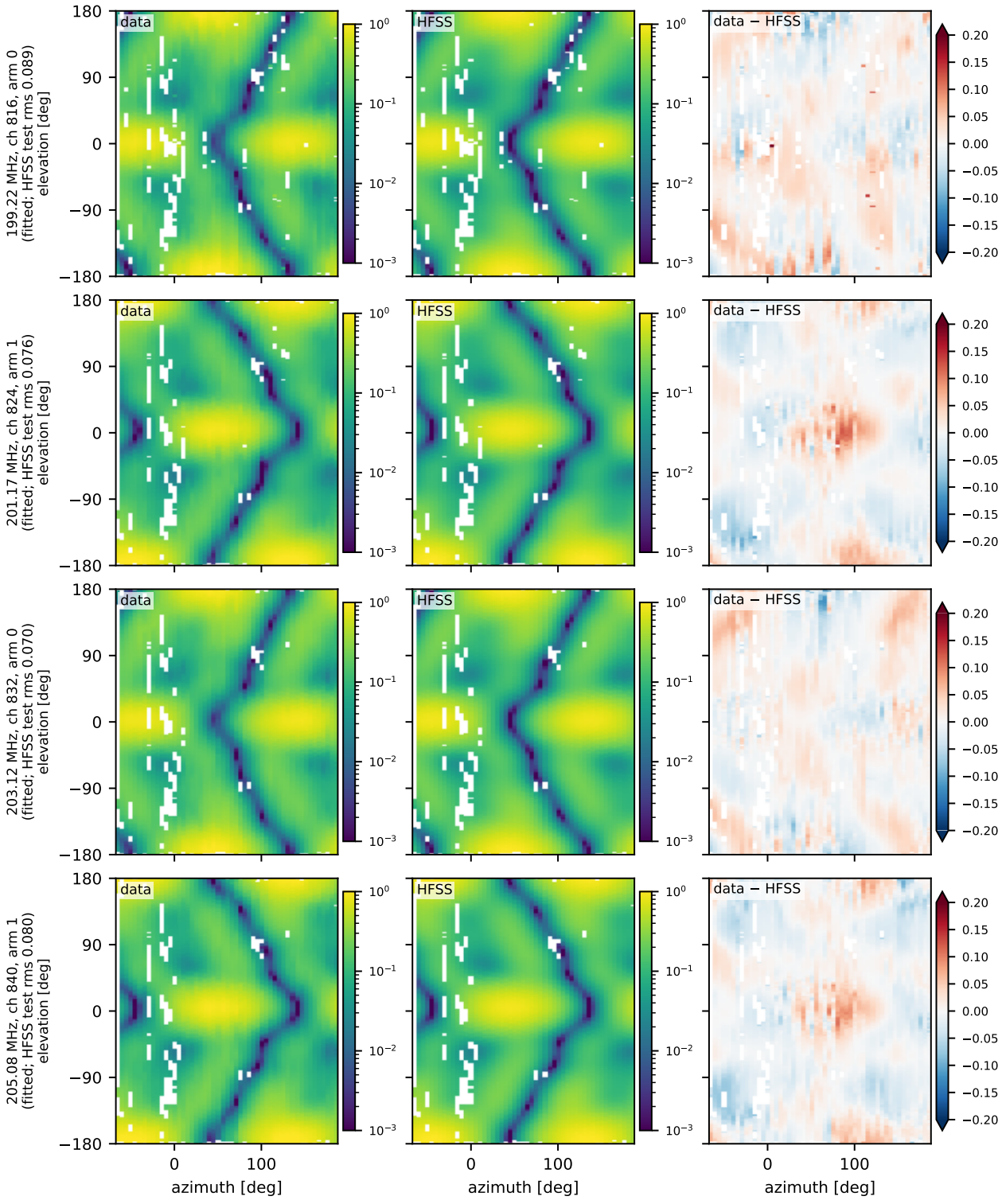


Figure 27: Raster maps, 199.22–205.08 MHz (page 20 of 26); panels as in Fig. 4.

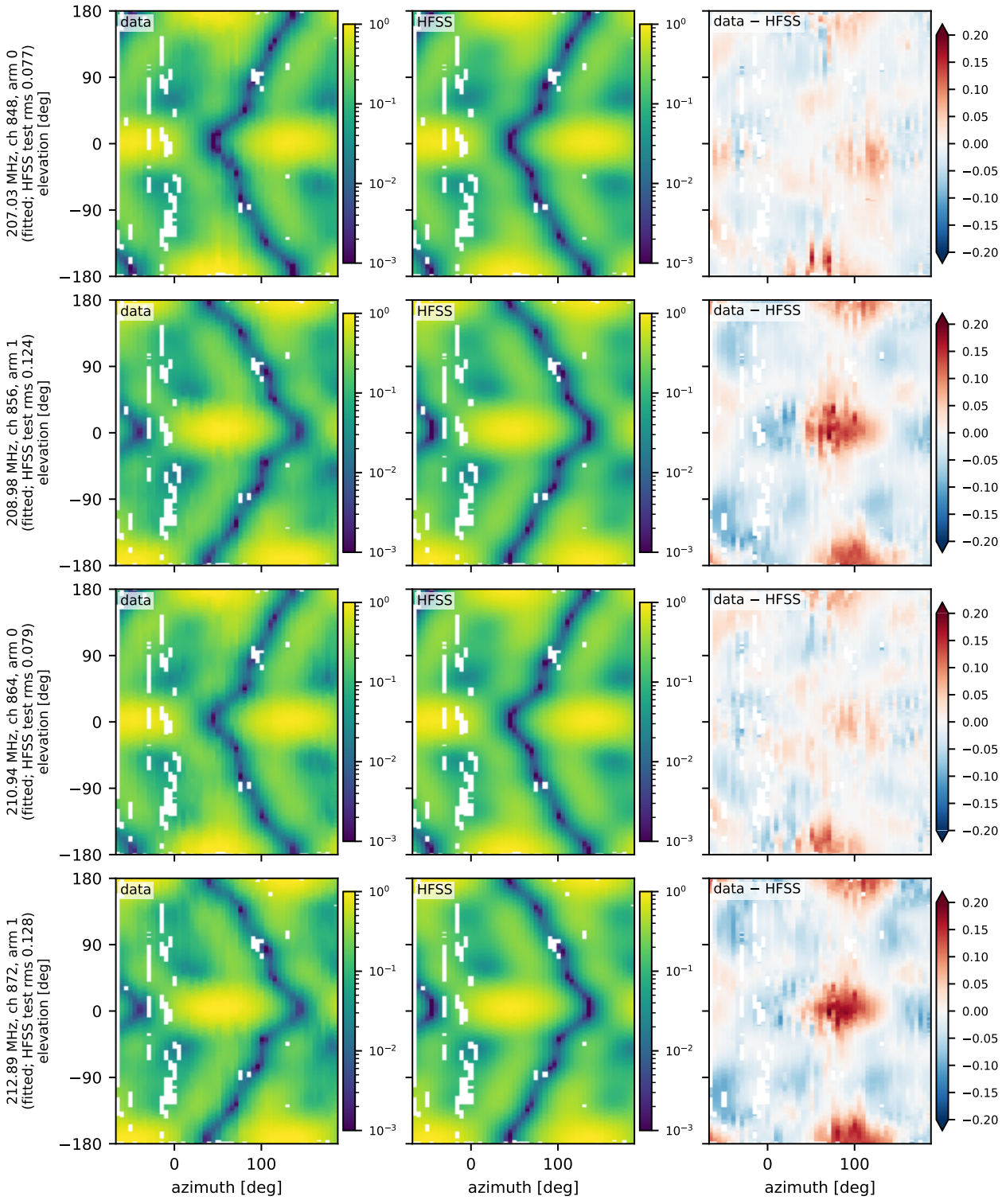


Figure 28: Raster maps, 207.03–212.89 MHz (page 21 of 26); panels as in Fig. 4.

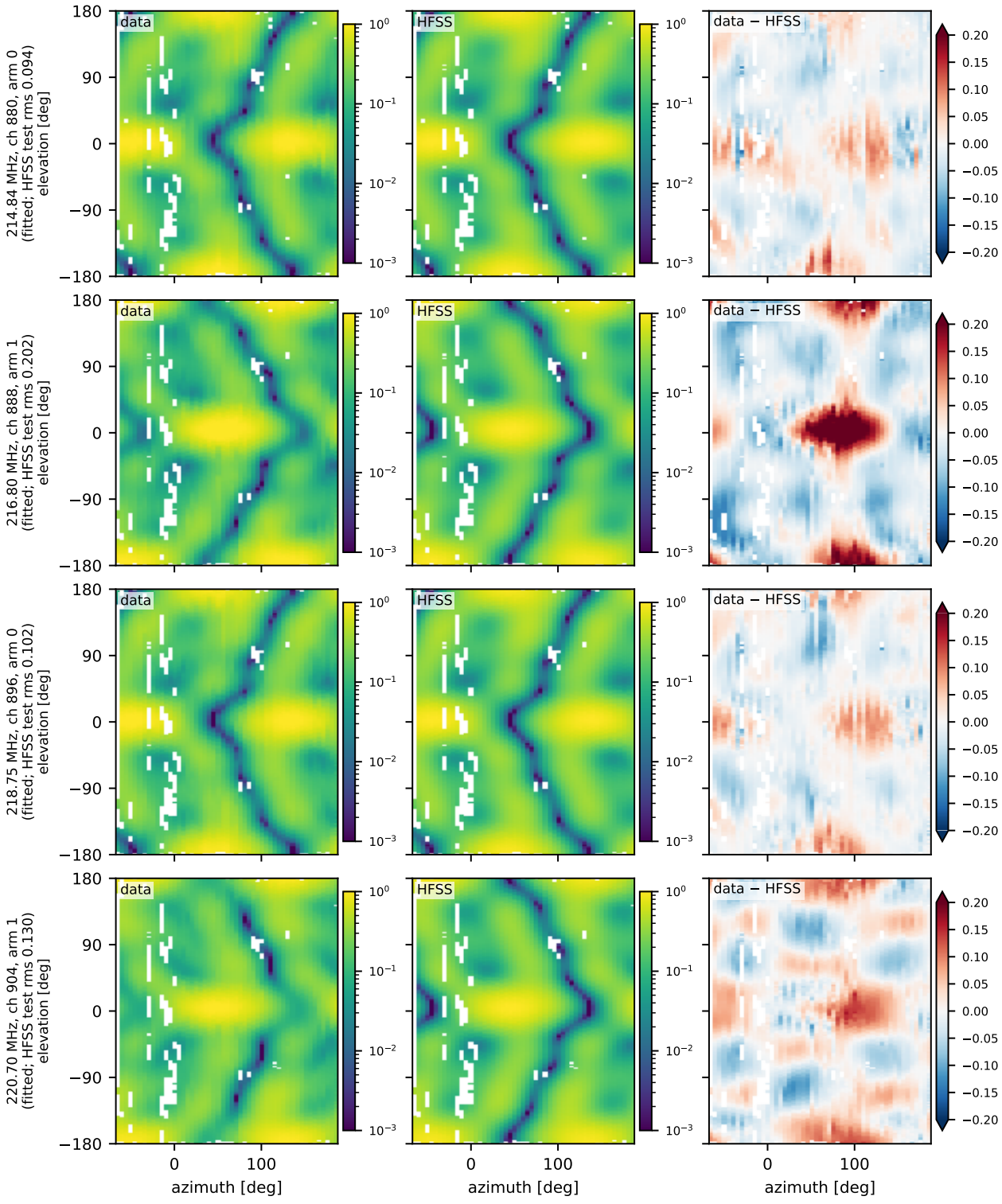


Figure 29: Raster maps, 214.84–220.70 MHz (page 22 of 26); panels as in Fig. 4.

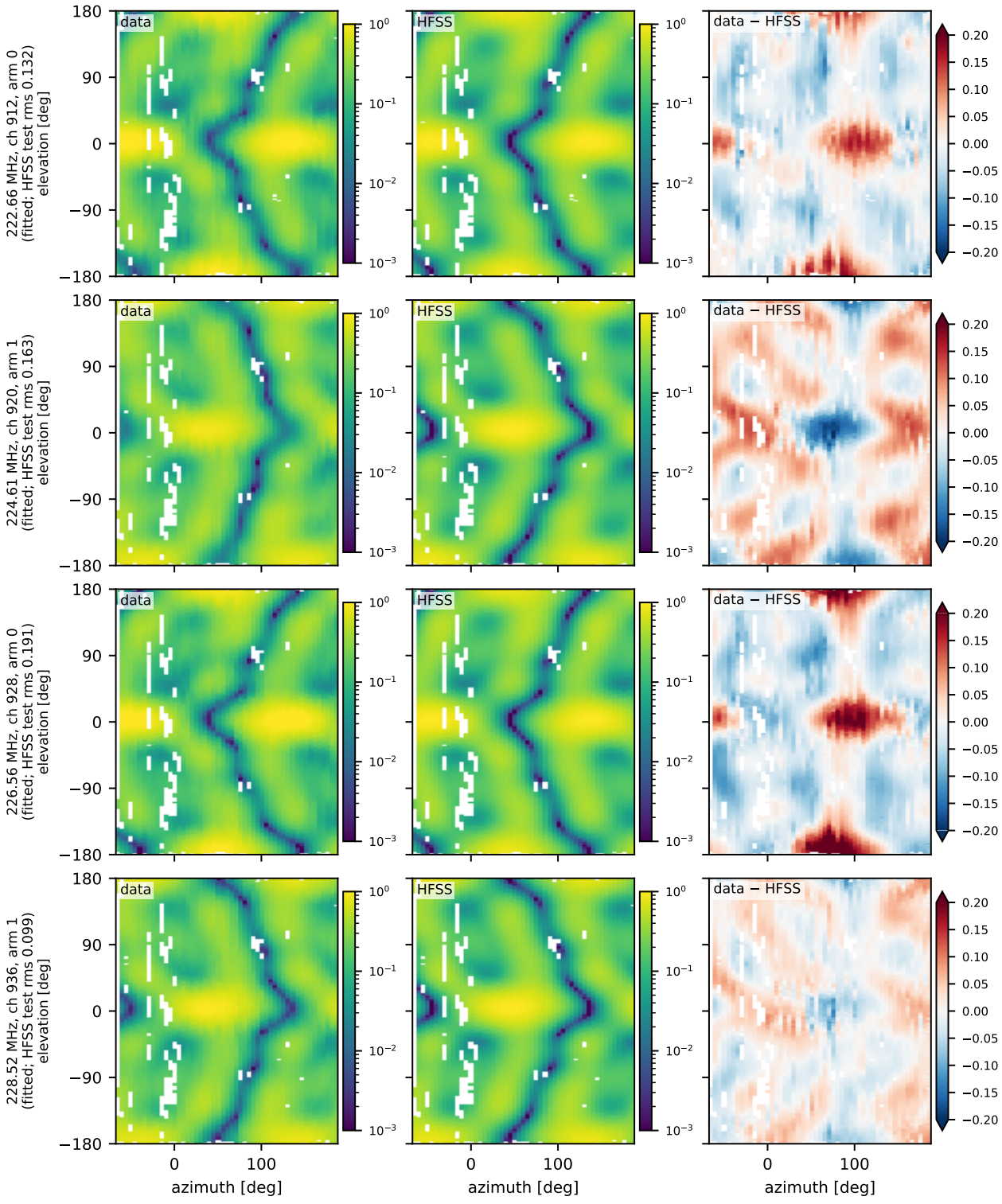


Figure 30: Raster maps, 222.66–228.52 MHz (page 23 of 26); panels as in Fig. 4.

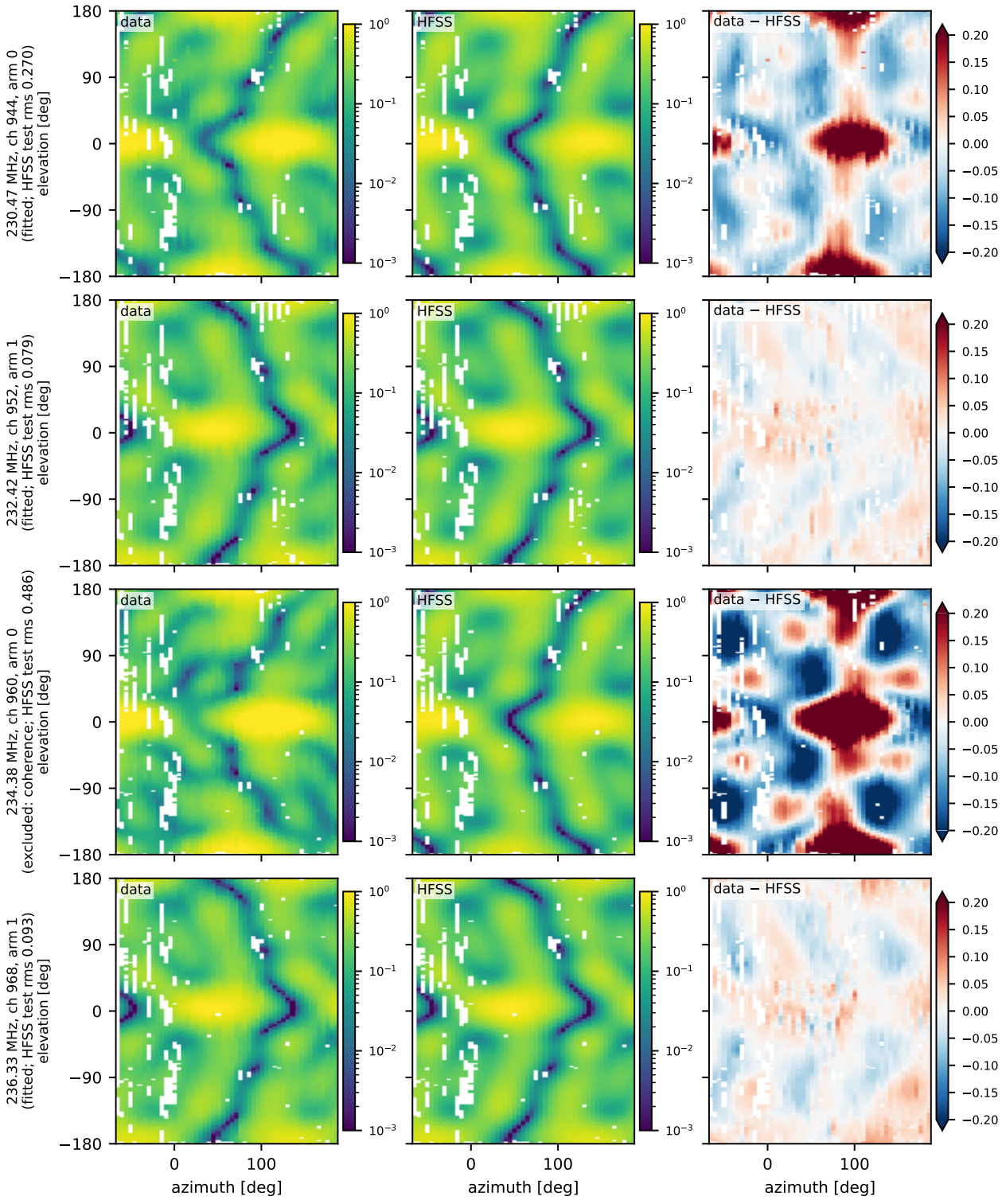


Figure 31: Raster maps, 230.47–236.33 MHz (page 24 of 26); panels as in Fig. 4.

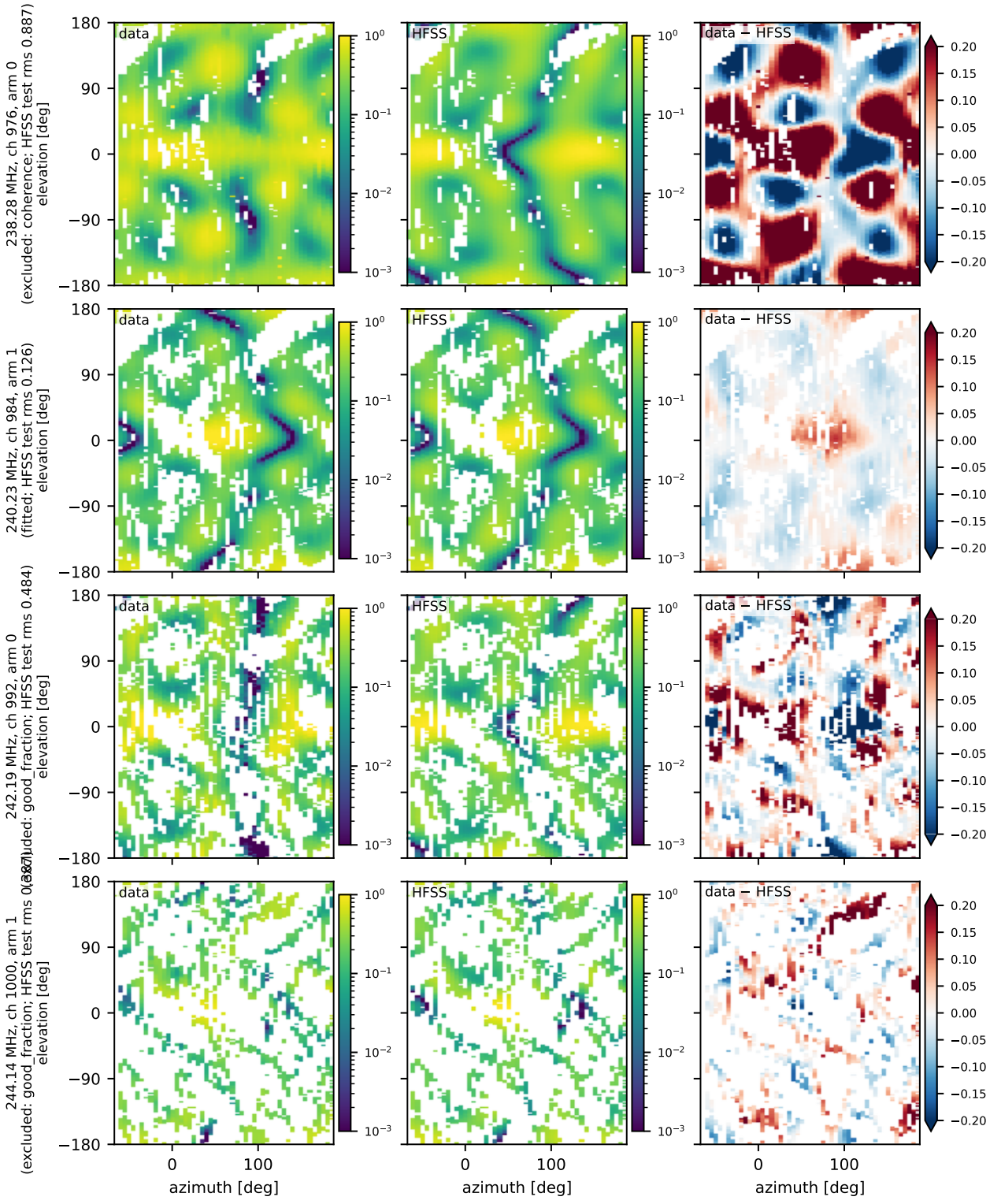


Figure 32: Raster maps, 238.28–244.14 MHz (page 25 of 26); panels as in Fig. 4.

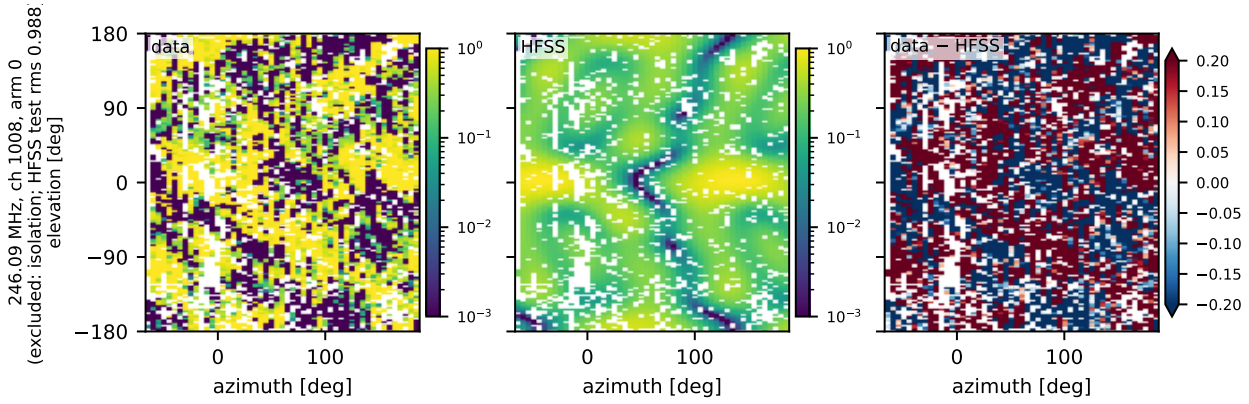


Figure 33: Raster maps, 246.09 MHz (page 26 of 26); panels as in Fig. 4.